TestBoard_ADC124 - upgraded firmware

14 October 2026 | No sensor strips connected | Charger connected | Production measurements

Scope and interpretation

Both full arrays: 50 useful channels/sweep. Manual timed acquisition; repeat 1; continuous-CS burst batches; requested SPI 16 MHz unless explicitly labeled 8 MHz. ADC mode scans address groups in array A, then B. MUX mode switches both banks before one shared settling wait, then reads ADC A and ADC B on their shared SPI bus.

Sensor delays 1, 2, 5, 10 and 50 us, with VMID delay explicitly equal to the sensor delay. Each has VMID off/on, both scan modes and all three engines. Extra tests: sensor 5 us with VMID on and VMID waits 0/1 us; sensor=VMID=5 us at 8 MHz with VMID off/on. Total 84 settings x 3 production repetitions, each 1 s warm-up + 15 s measurement. Diagnostic captures are separate and are excluded from production rate/noise results.

Nominal reference 3.3 V; 1 count = 3.3/4096 V = 0.805664 mV. Biases A: 470k / 1M / 470k / 1M; B: 1M / 470k / 1M / 470k. No rail or SPI-edge measurement was made. This report qualifies the disconnected setup; connected-sensor testing is a separate next phase.

Production captures	Diagnostic captures	Complete received sweeps	Accounted USB discards	Qualified quiet records
252	84	18,420,451	766	252/252

Firmware error totals: {'transfer_errors': 0, 'data_errors': 0, 'usb_write_errors': 0}. Every captured stream passed SHA-256, independent binary decoding, 12-bit code checks, wire-route validation, nonoverlapping device timestamps and sent/received accounting.

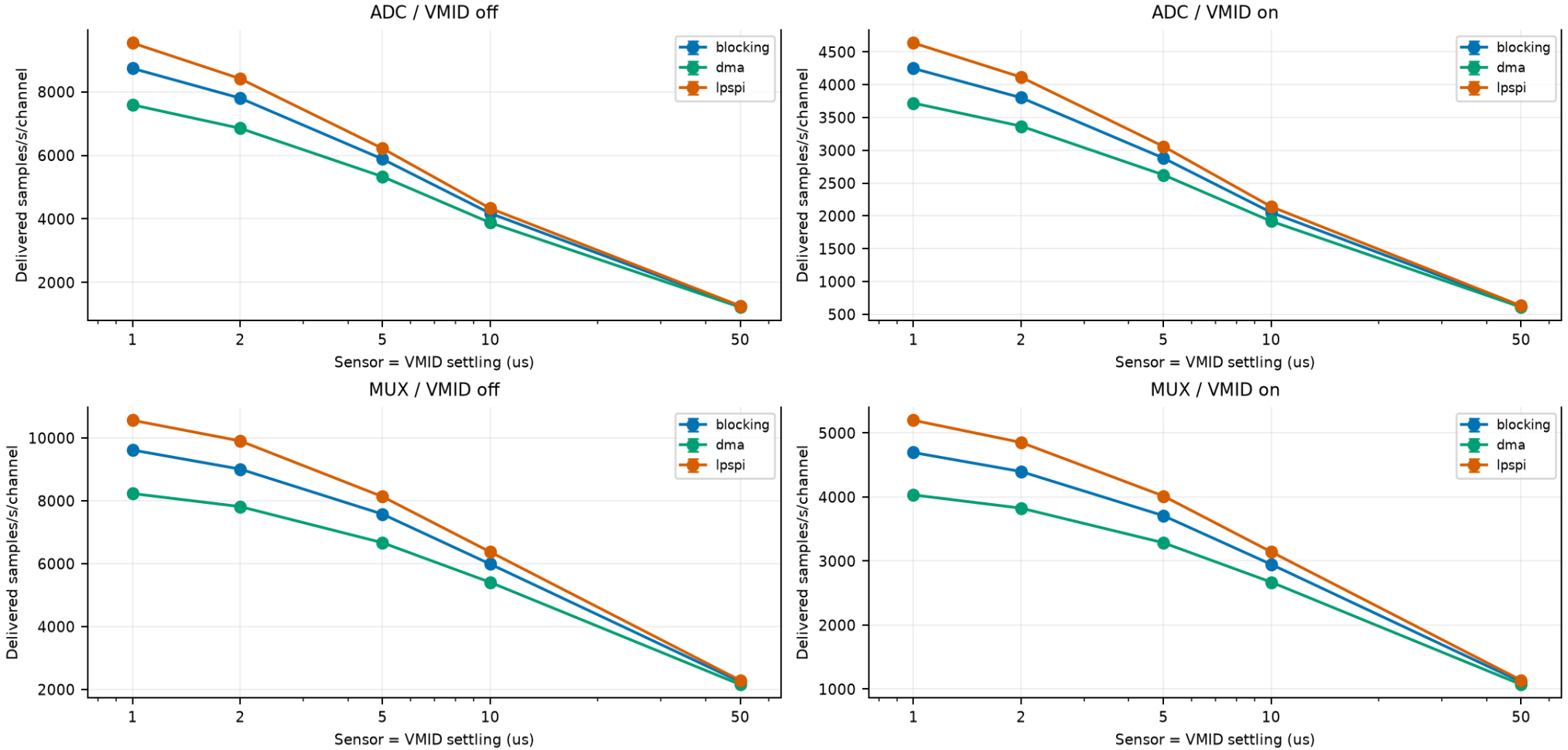
LPSPI / 5 us	VMID off rate	VMID on rate	VMID on worst SD (counts / mV)	VMID on worst p-p (counts / mV)
ADC	6,217.0	3,059.4	0.489 / 0.394	2.000 / 1.611
MUX	8,132.6	4,016.1	0.493 / 0.398	2.000 / 1.611

At 5 us, LPSPI rate gains versus yesterday's same-mode burst firmware: ADC off 9.42%; ADC on 9.80%; MUX off 13.86%; MUX on 15.26%.

Speed and noise must be assessed together: MUX/LPSPI at 5 us has worst retained sensor SD 0.497 / 0.401 counts / mV with VMID off, versus 0.493 / 0.398 with VMID on. Isolated channel variation remains in the analysis.

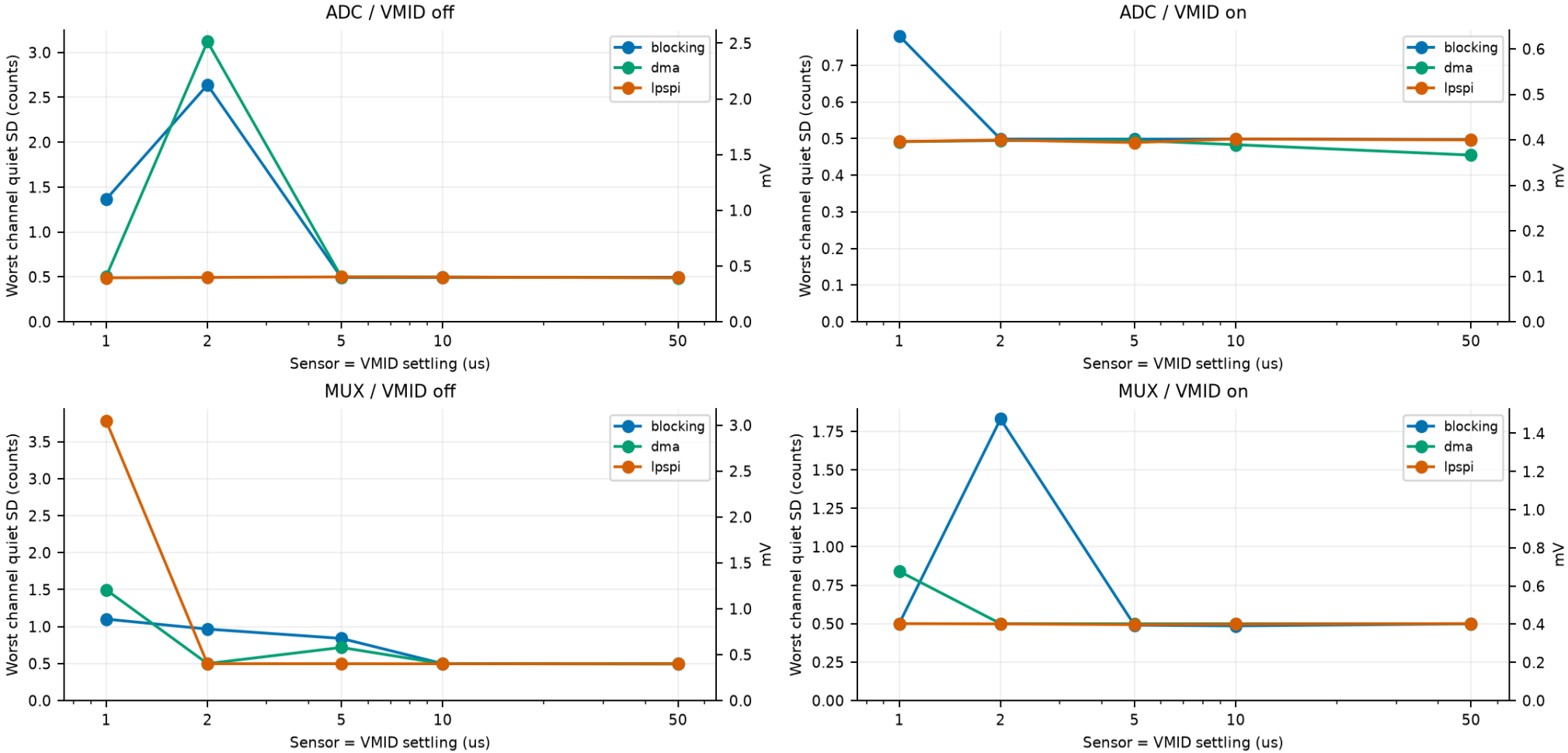
Primary baseline/noise values use between-burst windows: the earlier 50 ms mask plus rejection of separated, coherent short excursions. Broader-window noise is retained separately. The same two-stage method reanalyzes both historical datasets; its noise values can differ from their printed reports. Repetition summaries use medians; all route values, masks, ranges and comparisons are in CSV.

Sampling rate - both scan modes and all engines



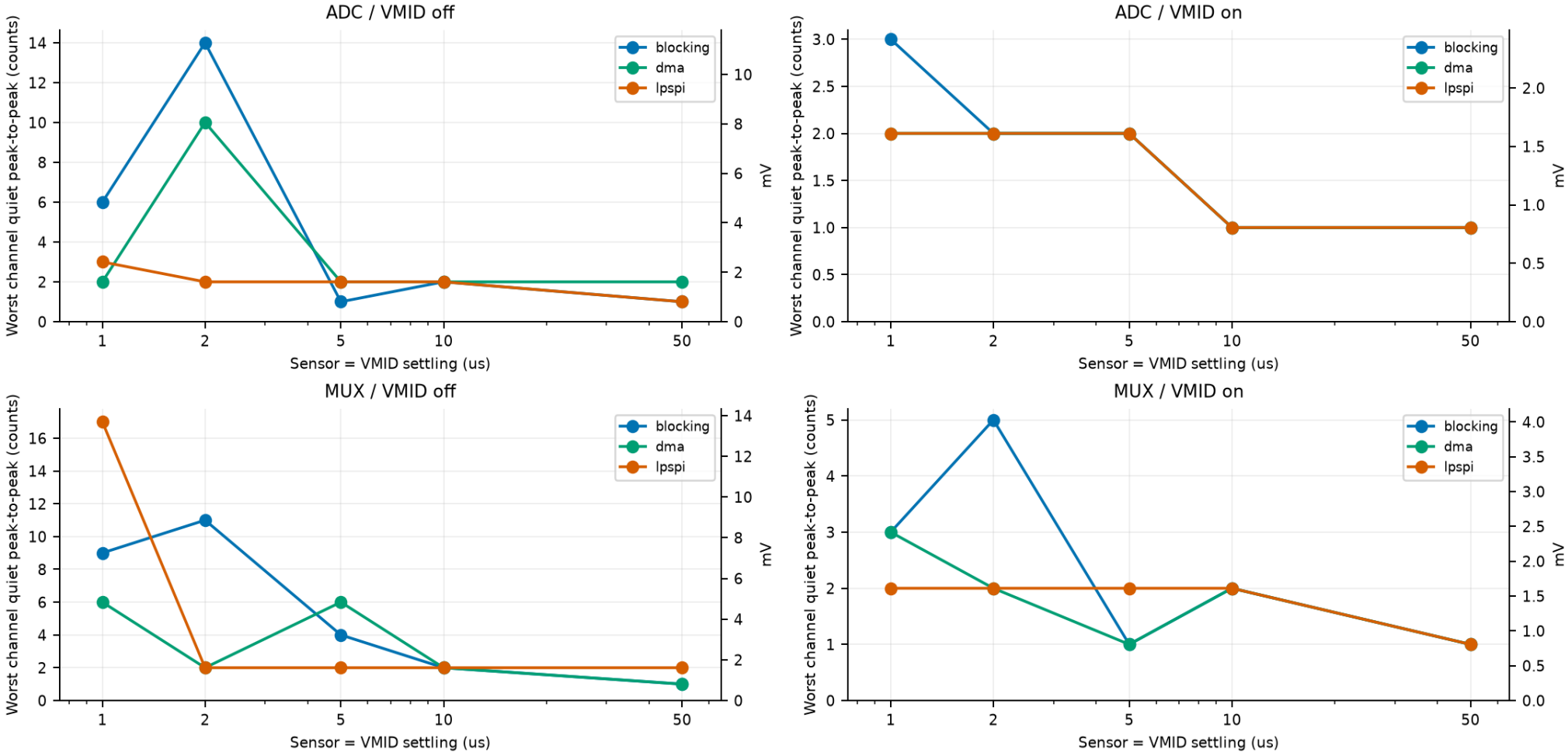
One sweep contains one value for each selected channel; sweep rate equals per-channel sampling rate. Multiply by 50 for useful aggregate values/s. Error bars show the minimum/maximum of three repetition rates. USB discards can lower delivered rate relative to sampling.

Worst channel quiet SD - absolute levels



16 MHz. For each physical route, combine qualified repetition metrics by median; show the maximum across all 50 routes. No individual sample clipping. Per-route values and repetition ranges are in the channel CSV.

Worst channel quiet peak-to-peak - absolute levels



16 MHz. For each physical route, combine qualified repetition metrics by median; show the maximum across all 50 routes. No individual sample clipping. Per-route values and repetition ranges are in the channel CSV.

ADC scan / VMID off - rate, quiet noise and baseline spread

16 MHz; sensor = VMID settling. Noise and spread columns show counts / mV. Baseline spread is max-min across the 25 per-route median baselines in that array. SD is temporal sample SD around each channel's own mean; it is separate from baseline variation across channels.

Engine	Delay us	Samples/s/ch	Median SD c/mV	Worst SD c/mV	Worst p-p c/mV	A spread c/mV	B spread c/mV	Quiet s range
blocking	1	8,735.4	0.204 / 0.164	1.363 / 1.098	6.000 / 4.834	84.000 / 67.676	188.000 / 151.465	5.8-14.9
blocking	2	7,800.0	0.185 / 0.149	2.638 / 2.125	14.000 / 11.279	83.000 / 66.870	167.000 / 134.546	15.0-15.0
blocking	5	5,882.4	0.092 / 0.074	0.492 / 0.396	1.000 / 0.806	79.000 / 63.647	125.000 / 100.708	15.0-15.0
blocking	10	4,176.4	0.028 / 0.023	0.492 / 0.396	2.000 / 1.611	74.000 / 59.619	86.000 / 69.287	15.0-15.0
blocking	50	1,250.9	0.036 / 0.029	0.495 / 0.399	1.000 / 0.806	41.000 / 33.032	12.000 / 9.668	15.0-15.0
dma	1	7,588.8	0.102 / 0.083	0.503 / 0.405	2.000 / 1.611	83.000 / 66.870	161.000 / 129.712	15.0-15.0
dma	2	6,850.3	0.177 / 0.143	3.120 / 2.514	10.000 / 8.057	82.000 / 66.064	145.000 / 116.821	15.0-15.0
dma	5	5,329.9	0.090 / 0.073	0.498 / 0.401	2.000 / 1.611	78.000 / 62.842	111.000 / 89.429	15.0-15.0
dma	10	3,880.6	0.030 / 0.025	0.498 / 0.401	2.000 / 1.611	73.000 / 58.813	79.000 / 63.647	15.0-15.0
dma	50	1,223.1	0.042 / 0.034	0.490 / 0.395	2.000 / 1.611	41.000 / 33.032	11.000 / 8.862	15.0-15.0
lpspi	1	9,534.6	0.113 / 0.091	0.489 / 0.394	3.000 / 2.417	85.000 / 68.481	199.000 / 160.327	15.0-15.0
lpspi	2	8,412.4	0.060 / 0.048	0.494 / 0.398	2.000 / 1.611	84.000 / 67.676	176.000 / 141.797	15.0-15.0
lpspi	5	6,217.0	0.047 / 0.038	0.498 / 0.402	2.000 / 1.611	80.000 / 64.453	128.000 / 103.125	15.0-15.0
lpspi	10	4,332.0	0.105 / 0.085	0.497 / 0.400	2.000 / 1.611	74.000 / 59.619	87.000 / 70.093	15.0-15.0
lpspi	50	1,264.5	0.031 / 0.025	0.490 / 0.395	1.000 / 0.806	42.000 / 33.838	12.000 / 9.668	15.0-15.0

Full voltage values are saved in volts in summary.csv and channel_summary.csv. Peak-to-peak is the actual retained maximum minus minimum, without individual sample clipping. It depends on retained observation length and sampling rate.

ADC scan / VMID on - rate, quiet noise and baseline spread

16 MHz; sensor = VMID settling. Noise and spread columns show counts / mV. Baseline spread is max-min across the 25 per-route median baselines in that array. SD is temporal sample SD around each channel's own mean; it is separate from baseline variation across channels.

Engine	Delay us	Samples/s/ch	Median SD c/mV	Worst SD c/mV	Worst p-p c/mV	A spread c/mV	B spread c/mV	Quiet s range
blocking	1	4,247.9	0.088 / 0.071	0.780 / 0.629	3.000 / 2.417	34.000 / 27.393	87.000 / 70.093	15.0-15.0
blocking	2	3,802.3	0.097 / 0.078	0.498 / 0.402	2.000 / 1.611	33.000 / 26.587	84.000 / 67.676	15.0-15.0
blocking	5	2,881.8	0.113 / 0.091	0.498 / 0.401	2.000 / 1.611	34.000 / 27.393	75.000 / 60.425	15.0-15.0
blocking	10	2,053.9	0.144 / 0.116	0.499 / 0.402	1.000 / 0.806	35.000 / 28.198	64.000 / 51.562	15.0-15.0
blocking	50	622.4	0.112 / 0.090	0.497 / 0.401	1.000 / 0.806	27.000 / 21.753	12.000 / 9.668	15.0-15.0
dma	1	3,719.2	0.123 / 0.099	0.491 / 0.396	2.000 / 1.611	34.000 / 27.393	81.000 / 65.259	15.0-15.0
dma	2	3,368.4	0.158 / 0.127	0.495 / 0.398	2.000 / 1.611	34.000 / 27.393	77.000 / 62.036	15.0-15.0
dma	5	2,626.0	0.138 / 0.111	0.495 / 0.399	2.000 / 1.611	35.000 / 28.198	70.000 / 56.396	15.0-15.0
dma	10	1,919.6	0.075 / 0.060	0.483 / 0.389	1.000 / 0.806	35.000 / 28.198	60.000 / 48.340	15.0-15.0
dma	50	609.5	0.032 / 0.026	0.455 / 0.366	1.000 / 0.806	27.000 / 21.753	11.000 / 8.862	15.0-15.0
lpspi	1	4,636.4	0.088 / 0.071	0.492 / 0.396	2.000 / 1.611	33.000 / 26.587	84.000 / 67.676	15.0-15.0
lpspi	2	4,115.2	0.117 / 0.094	0.496 / 0.399	2.000 / 1.611	34.000 / 27.393	81.000 / 65.259	15.0-15.0
lpspi	5	3,059.4	0.049 / 0.039	0.489 / 0.394	2.000 / 1.611	34.000 / 27.393	74.000 / 59.619	15.0-15.0
lpspi	10	2,142.0	0.107 / 0.086	0.499 / 0.402	1.000 / 0.806	34.000 / 27.393	63.000 / 50.757	15.0-15.0
lpspi	50	630.2	0.073 / 0.059	0.496 / 0.400	1.000 / 0.806	28.000 / 22.559	12.000 / 9.668	15.0-15.0

Full voltage values are saved in volts in summary.csv and channel_summary.csv. Peak-to-peak is the actual retained maximum minus minimum, without individual sample clipping. It depends on retained observation length and sampling rate.

MUX scan / VMID off - rate, quiet noise and baseline spread

16 MHz; sensor = VMID settling. Noise and spread columns show counts / mV. Baseline spread is max-min across the 25 per-route median baselines in that array. SD is temporal sample SD around each channel's own mean; it is separate from baseline variation across channels.

Engine	Delay us	Samples/s/ch	Median SD c/mV	Worst SD c/mV	Worst p-p c/mV	A spread c/mV	B spread c/mV	Quiet s range
blocking	1	9,614.8	0.069 / 0.056	1.101 / 0.887	9.000 / 7.251	79.000 / 63.647	97.000 / 78.149	15.0-15.0
blocking	2	9,008.0	0.091 / 0.073	0.966 / 0.778	11.000 / 8.862	78.000 / 62.842	90.000 / 72.510	15.0-15.0
blocking	5	7,575.8	0.051 / 0.041	0.840 / 0.677	4.000 / 3.223	75.000 / 60.425	75.000 / 60.425	15.0-15.0
blocking	10	5,988.0	0.057 / 0.046	0.498 / 0.401	2.000 / 1.611	70.000 / 56.396	56.000 / 45.117	15.0-15.0
blocking	50	2,241.4	0.027 / 0.021	0.499 / 0.402	1.000 / 0.806	40.000 / 32.227	8.000 / 6.445	15.0-15.0
dma	1	8,229.9	0.063 / 0.050	1.498 / 1.207	6.000 / 4.834	79.000 / 63.647	82.000 / 66.064	15.0-15.0
dma	2	7,815.5	0.064 / 0.051	0.499 / 0.402	2.000 / 1.611	78.000 / 62.842	77.000 / 62.036	15.0-15.0
dma	5	6,666.3	0.049 / 0.040	0.717 / 0.578	6.000 / 4.834	74.000 / 59.619	65.000 / 52.368	15.0-15.0
dma	10	5,407.2	0.065 / 0.052	0.497 / 0.401	2.000 / 1.611	69.000 / 55.591	49.000 / 39.478	15.0-15.0
dma	50	2,155.7	0.043 / 0.035	0.493 / 0.397	1.000 / 0.806	40.000 / 32.227	8.000 / 6.445	15.0-15.0
lpspi	1	10,561.5	0.188 / 0.151	3.782 / 3.047	17.000 / 13.696	80.000 / 64.453	101.000 / 81.372	15.0-15.0
lpspi	2	9,904.7	0.085 / 0.069	0.498 / 0.401	2.000 / 1.611	79.000 / 63.647	97.000 / 78.149	15.0-15.0
lpspi	5	8,132.6	0.080 / 0.065	0.497 / 0.401	2.000 / 1.611	76.000 / 61.230	79.000 / 63.647	4.8-4.8
lpspi	10	6,374.0	0.025 / 0.020	0.497 / 0.400	2.000 / 1.611	71.000 / 57.202	59.000 / 47.534	15.0-15.0
lpspi	50	2,288.9	0.020 / 0.016	0.497 / 0.400	2.000 / 1.611	40.000 / 32.227	9.000 / 7.251	15.0-15.0

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MUX scan / VMID on - rate, quiet noise and baseline spread

16 MHz; sensor = VMID settling. Noise and spread columns show counts / mV. Baseline spread is max-min across the 25 per-route median baselines in that array. SD is temporal sample SD around each channel's own mean; it is separate from baseline variation across channels.

Engine	Delay us	Samples/s/ch	Median SD c/mV	Worst SD c/mV	Worst p-p c/mV	A spread c/mV	B spread c/mV	Quiet s range
blocking	1	4,694.4	0.138 / 0.111	0.500 / 0.403	3.000 / 2.417	34.000 / 27.393	67.000 / 53.979	14.2-14.2
blocking	2	4,397.6	0.096 / 0.078	1.831 / 1.475	5.000 / 4.028	33.000 / 26.587	65.000 / 52.368	15.0-15.0
blocking	5	3,708.3	0.080 / 0.065	0.490 / 0.395	1.000 / 0.806	34.000 / 27.393	57.000 / 45.923	15.0-15.0
blocking	10	2,948.8	0.093 / 0.075	0.486 / 0.391	2.000 / 1.611	34.000 / 27.393	47.000 / 37.866	15.0-15.0
blocking	50	1,112.6	0.169 / 0.136	0.499 / 0.402	1.000 / 0.806	27.000 / 21.753	8.000 / 6.445	15.0-15.0
dma	1	4,032.3	0.146 / 0.118	0.840 / 0.677	3.000 / 2.417	34.000 / 27.393	61.000 / 49.146	15.0-15.0
dma	2	3,825.9	0.107 / 0.086	0.500 / 0.403	2.000 / 1.611	34.000 / 27.393	58.000 / 46.729	15.0-15.0
dma	5	3,285.4	0.060 / 0.048	0.499 / 0.402	1.000 / 0.806	34.000 / 27.393	52.000 / 41.895	15.0-15.0
dma	10	2,671.1	0.091 / 0.074	0.499 / 0.402	2.000 / 1.611	35.000 / 28.198	42.000 / 33.838	15.0-15.0
dma	50	1,071.4	0.044 / 0.035	0.499 / 0.402	1.000 / 0.806	27.000 / 21.753	7.000 / 5.640	15.0-15.0
lpspi	1	5,198.4	0.175 / 0.141	0.500 / 0.403	2.000 / 1.611	34.000 / 27.393	68.000 / 54.785	15.0-15.0
lpspi	2	4,852.4	0.069 / 0.056	0.498 / 0.401	2.000 / 1.611	34.000 / 27.393	66.000 / 53.174	15.0-15.0
lpspi	5	4,016.1	0.125 / 0.101	0.493 / 0.398	2.000 / 1.611	35.000 / 28.198	59.000 / 47.534	15.0-15.0
lpspi	10	3,144.3	0.115 / 0.093	0.497 / 0.400	2.000 / 1.611	35.000 / 28.198	49.000 / 39.478	15.0-15.0
lpspi	50	1,138.9	0.047 / 0.038	0.500 / 0.403	1.000 / 0.806	27.000 / 21.753	9.000 / 7.251	15.0-15.0

Full voltage values are saved in volts in summary.csv and channel_summary.csv. Peak-to-peak is the actual retained maximum minus minimum, without individual sample clipping. It depends on retained observation length and sampling rate.

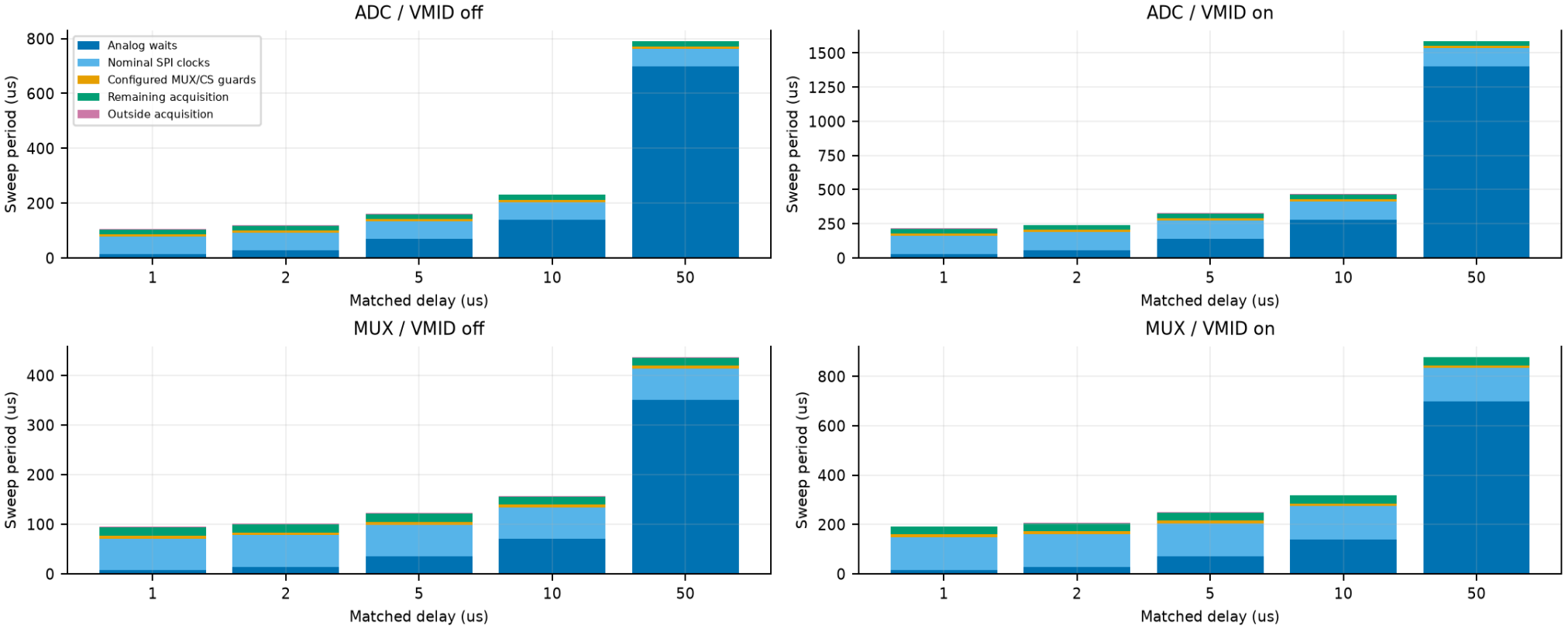
Between-burst noise and the retained broader-window measurement

LPSPI, sensor settling 5 us. The broader 50 ms method can include repeated submillisecond excursions. Primary noise excludes clearly separated excursions occurring coherently on at least two channels, with one whole-sweep guard on each side. The electrical source of these excursions was not independently measured.

Mode	VMID	VMID us	MHz	Broader SD c/mV	Between SD c/mV	Broader p-p c/mV	Between p-p c/mV	Extra rejected %	Quiet s range
ADC	off	5	16	0.498 / 0.402	0.498 / 0.402	2.000 / 1.611	2.000 / 1.611	0.00	15.00-15.00
ADC	on	5	16	0.489 / 0.394	0.489 / 0.394	2.000 / 1.611	2.000 / 1.611	0.00	15.00-15.00
ADC	on	0	16	0.499 / 0.402	0.499 / 0.402	2.000 / 1.611	2.000 / 1.611	0.00	14.97-14.97
ADC	on	1	16	0.491 / 0.395	0.491 / 0.395	2.000 / 1.611	2.000 / 1.611	0.00	15.00-15.00
ADC	off	5	8	0.495 / 0.399	0.495 / 0.399	1.000 / 0.806	1.000 / 0.806	0.00	14.99-14.99
ADC	on	5	8	0.501 / 0.403	0.501 / 0.403	2.000 / 1.611	2.000 / 1.611	0.00	15.00-15.00
MUX	off	5	16	1.433 / 1.155	0.497 / 0.401	7.000 / 5.640	2.000 / 1.611	67.83	4.82-4.84
MUX	on	5	16	0.493 / 0.398	0.493 / 0.398	2.000 / 1.611	2.000 / 1.611	0.00	15.00-15.00
MUX	on	0	16	0.815 / 0.657	0.815 / 0.657	3.000 / 2.417	3.000 / 2.417	0.00	14.98-14.98
MUX	on	1	16	0.500 / 0.403	0.500 / 0.403	2.000 / 1.611	2.000 / 1.611	0.00	15.00-15.00
MUX	off	5	8	0.684 / 0.551	0.684 / 0.551	3.000 / 2.417	3.000 / 2.417	0.00	15.00-15.00
MUX	on	5	8	0.670 / 0.540	0.670 / 0.540	3.000 / 2.417	3.000 / 2.417	0.00	15.00-15.00

Broader and between-burst metrics answer different questions. The broader metric retains burst burden; the conditional metric estimates noise between detected excursions. Detection sensitivity depends on amplitude and separation of the secondary state. It does not prove that every charger-related excursion was detected, or that reduced conditional SD alone means improved acquisition quality. Whole-sweep masks and unfiltered raw data remain available.

Timing breakdown - configured budgets and measured residuals



Mode / VMID	Prod period us	Prod acquisition us	Prod gap us	Diag prepare us	Diag transfer us	Diag encode us	Diag enqueue us
ADC/off	160.849	159.717	1.132	79.553	81.357	0.250	0.409
ADC/on	326.863	325.733	1.130	159.197	168.904	0.250	0.410
MUX/off	122.961	121.747	1.215	41.094	80.407	0.250	0.410
MUX/on	249.000	248.000	1.000	82.047	167.305	0.250	0.409

Diagnostic probes add overhead; these phase means are not subtracted from production totals. Clock and guard budgets are arithmetic estimates, not measured pin timings. ADC conversion overlaps its SPI clocks; do not add a separate conversion duration to the clock budget.

Timing by address, ADC channel and full sweep

Unit / fixed address	Useful values	Prime words	Total conversion words	Nominal SPI time at 16 / 8 MHz
One ADC, sensor address 0-5	4	1	5	5 / 10 us
One ADC, sensor address 6	1	1	2	2 / 4 us
Both ADCs, sensor address 0-5	8	2	10	10 / 20 us
Both ADCs, sensor address 6	2	2	4	4 / 8 us
One ADC, optional VMID batch	0	1	5	5 / 10 us
Both ADCs, optional VMID batches	0	2	10	10 / 20 us
Full sensor sweep, VMID off	50	14	64	64 / 128 us
Full sweep, VMID on	50	28	134	134 / 268 us

Each useful conversion occupies one 16-clock word: nominally 1 us at 16 MHz or 2 us at 8 MHz. This is conversion/clock time, not the elapsed time between revisits of that sensor. Each sensor is revisited once per complete sweep. A prime conversion is discarded at the start of each batch; actual preparation and batch software costs are additional.

ADC mode switches/settles 14 sensor bank-address groups; MUX mode switches the same 14 banks in 7 shared phases. VMID on adds 14 bank switches / 14 waits in ADC mode, or 14 bank switches / 7 waits in MUX mode. Coordinated switch guards cost 0.4 us per changed phase; CS setup/high allowances cost 0.2 us per burst batch. EN changes only for required address transitions. These are source-derived estimates, not oscilloscope edge measurements.

Mode	Engine	VMID	Measured intercept us	Measured slope us/us	Expected wait slope	Fit residual max us
ADC	blocking	off	100.125	13.9843	14	0.528
ADC	blocking	on	207.112	27.9930	28	0.306
ADC	dma	off	117.789	13.9960	14	0.198
ADC	dma	on	240.878	27.9988	28	0.076
ADC	lpspi	off	90.864	13.9996	14	0.018
ADC	lpspi	on	187.152	27.9927	28	0.542
MUX	blocking	off	97.082	6.9818	7	0.100
MUX	blocking	on	199.343	13.9896	14	0.375
MUX	dma	off	114.633	6.9872	7	0.657
MUX	dma	on	234.065	13.9873	14	0.661
MUX	lpspi	off	87.437	6.9882	7	0.584
MUX	lpspi	on	178.409	13.9923	14	0.630

Fit: period = intercept + slope x matched sensor/VMID delay. The intercept estimates total zero-configured-delay period; it cannot separately identify GPIO, per-word and per-batch costs. Details: delay_fits.csv.

Baseline differences between the four ADC input lanes

LPSPI; sensor = VMID settling = 5 us; 16 MHz. Each ADC input lane corresponds to one external MUX. Lane center is the median of that lane's 6 or 7 channel baselines. The spread across centers compares the four input groups. These are descriptive offsets in this disconnected setup.

Mode	VMID	Array	MUX1 center c / V	MUX2 center c / V	MUX3 center c / V	MUX4 center c / V	Lane-center spread c/mV
ADC	off	A	2,026.0 / 1.632275	1,997.5 / 1.609314	2,022.0 / 1.629053	2,047.0 / 1.649194	49.500 / 39.880
ADC	off	B	1,885.5 / 1.519080	1,986.5 / 1.600452	1,903.5 / 1.533582	1,999.0 / 1.610522	113.500 / 91.443
ADC	on	A	1,998.0 / 1.609717	1,974.5 / 1.590784	1,986.5 / 1.600452	1,994.0 / 1.606494	23.500 / 18.933
ADC	on	B	1,939.0 / 1.562183	1,995.5 / 1.607703	1,951.5 / 1.572253	2,006.0 / 1.616162	67.000 / 53.979
MUX	off	A	2,034.0 / 1.638721	2,008.5 / 1.618176	2,031.5 / 1.636707	2,045.0 / 1.647583	36.500 / 29.407
MUX	off	B	1,958.5 / 1.577893	2,024.0 / 1.630664	1,968.0 / 1.585547	2,027.0 / 1.633081	68.500 / 55.188
MUX	on	A	1,999.0 / 1.610522	1,975.0 / 1.591187	1,988.0 / 1.601660	1,995.0 / 1.607300	24.000 / 19.336
MUX	on	B	1,979.0 / 1.594409	2,026.0 / 1.632275	1,986.0 / 1.600049	2,029.0 / 1.634692	50.000 / 40.283

mux_group_summary.csv contains every engine, delay and experiment. address_baseline_spread.csv compares ADC channels at each common external address: four lanes at inputs 0-5; input 6 has one populated lane, so its single-value spread is zero and does not measure inter-lane variation. The channel line graphs preserve every individual route.

Independent VMID wait - sensor settling fixed at 5 us

16 MHz; VMID on. Only the extra VMID analog wait changes. All 134 SPI conversion words, 28 burst batches, selector primes and switching guards remain.

Mode	Engine	VMID us	Rate /ch	Gain % vs5	Saved us measured	Saved us estimated	Worst SD c/mV	Worst p-p c/mV	A spread c/mV	B spread c/mV
ADC	blocking	0	3,615.8	25.47	70.432	70	0.499 / 0.402	2.000 / 1.611	34.000 / 27.393	74.000 / 59.619
ADC	blocking	1	3,432.1	19.09	55.631	56	0.816 / 0.657	2.000 / 1.611	34.000 / 27.393	74.000 / 59.619
ADC	blocking	5	2,881.8	0.00	0.000	0	0.498 / 0.401	2.000 / 1.611	34.000 / 27.393	75.000 / 60.425
ADC	dma	0	3,215.4	22.45	69.809	70	0.500 / 0.403	2.000 / 1.611	35.000 / 28.198	69.000 / 55.591
ADC	dma	1	3,078.7	17.24	56.001	56	0.495 / 0.399	2.000 / 1.611	35.000 / 28.198	69.000 / 55.591
ADC	dma	5	2,626.0	0.00	0.000	0	0.495 / 0.399	2.000 / 1.611	35.000 / 28.198	70.000 / 56.396
ADC	lpspi	0	3,899.5	27.46	70.419	70	0.499 / 0.402	2.000 / 1.611	35.000 / 28.198	73.000 / 58.813
ADC	lpspi	1	3,678.1	20.22	54.982	56	0.491 / 0.395	2.000 / 1.611	35.000 / 28.198	73.000 / 58.813
ADC	lpspi	5	3,059.4	0.00	0.000	0	0.489 / 0.394	2.000 / 1.611	34.000 / 27.393	74.000 / 59.619
MUX	blocking	0	4,284.9	15.55	36.289	35	0.517 / 0.417	2.000 / 1.611	34.000 / 27.393	57.000 / 45.923
MUX	blocking	1	4,152.5	11.98	28.849	28	0.498 / 0.401	2.000 / 1.611	34.000 / 27.393	57.000 / 45.923
MUX	blocking	5	3,708.3	0.00	0.000	0	0.490 / 0.395	1.000 / 0.806	34.000 / 27.393	57.000 / 45.923
MUX	dma	0	3,729.3	13.51	36.234	35	0.495 / 0.399	2.000 / 1.611	35.000 / 28.198	51.000 / 41.089
MUX	dma	1	3,618.2	10.13	28.000	28	0.498 / 0.401	2.000 / 1.611	35.000 / 28.198	51.000 / 41.089
MUX	dma	5	3,285.4	0.00	0.000	0	0.499 / 0.402	1.000 / 0.806	34.000 / 27.393	52.000 / 41.895
MUX	lpspi	0	4,694.8	16.90	36.000	35	0.815 / 0.657	3.000 / 2.417	35.000 / 28.198	58.000 / 46.729
MUX	lpspi	1	4,545.5	13.18	29.000	28	0.500 / 0.403	2.000 / 1.611	35.000 / 28.198	58.000 / 46.729
MUX	lpspi	5	4,016.1	0.00	0.000	0	0.493 / 0.398	2.000 / 1.611	35.000 / 28.198	59.000 / 47.534

VMID zero skips the configured analog wait, retaining mandatory switch guards and discarded VMID conversions. The transmitted payload contains sensor readings; discarded VMID responses themselves are not recorded or assessed for analog noise.

SPI clock comparison - 8 MHz versus 16 MHz

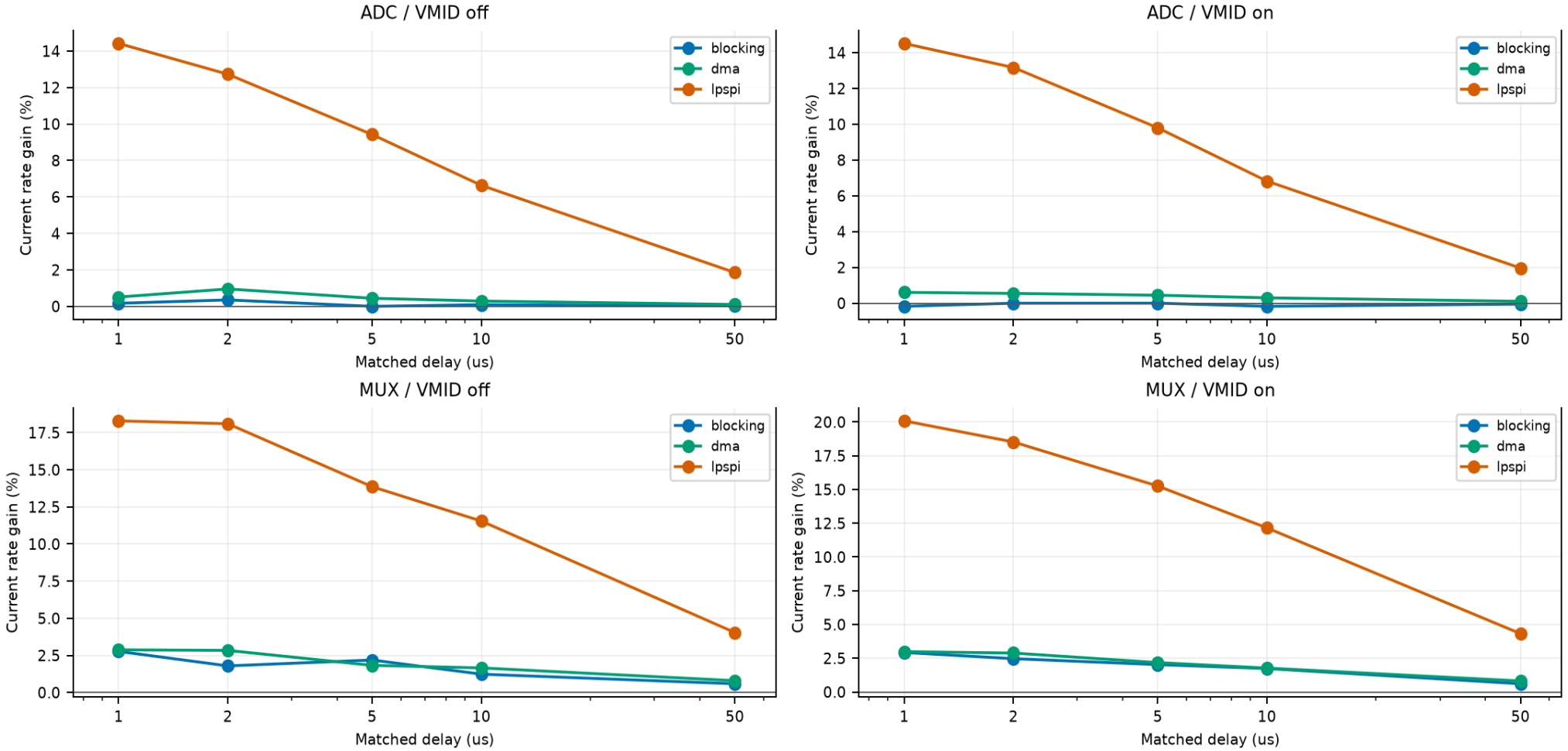
Sensor settling = VMID settling = 5 us. Both VMID off/on are tested. Requested clocks are reported; actual SCLK was not probed. Slower clocks also change ADC acquisition timing and analog history.

Mode	Engine	VMID	8 MHz rate	16 MHz rate	16 MHz gain %	8 MHz SD c/mV	16 MHz SD c/mV	8 MHz p-p c/mV	16 MHz p-p c/mV
ADC	blocking	off	4,245.0	5,882.4	38.57	0.498 / 0.401	0.492 / 0.396	1.000 / 0.806	1.000 / 0.806
ADC	blocking	on	2,060.4	2,881.8	39.87	0.499 / 0.402	0.498 / 0.401	2.000 / 1.611	2.000 / 1.611
ADC	dma	off	3,907.7	5,329.9	36.39	0.496 / 0.400	0.498 / 0.401	2.000 / 1.611	2.000 / 1.611
ADC	dma	on	1,909.4	2,626.0	37.53	0.482 / 0.388	0.495 / 0.399	2.000 / 1.611	2.000 / 1.611
ADC	lpspi	off	4,401.5	6,217.0	41.25	0.495 / 0.399	0.498 / 0.402	1.000 / 0.806	2.000 / 1.611
ADC	lpspi	on	2,147.9	3,059.4	42.44	0.501 / 0.403	0.489 / 0.394	2.000 / 1.611	2.000 / 1.611
MUX	blocking	off	5,057.0	7,575.8	49.81	1.432 / 1.154	0.840 / 0.677	4.000 / 3.223	4.000 / 3.223
MUX	blocking	on	2,451.0	3,708.3	51.30	1.073 / 0.865	0.490 / 0.395	4.000 / 3.223	1.000 / 0.806
MUX	dma	off	4,597.3	6,666.3	45.01	0.968 / 0.780	0.717 / 0.578	4.000 / 3.223	6.000 / 4.834
MUX	dma	on	2,239.2	3,285.4	46.72	0.977 / 0.787	0.499 / 0.402	3.000 / 2.417	1.000 / 0.806
MUX	lpspi	off	5,299.7	8,132.6	53.46	0.684 / 0.551	0.497 / 0.401	3.000 / 2.417	2.000 / 1.611
MUX	lpspi	on	2,579.5	4,016.1	55.69	0.670 / 0.540	0.493 / 0.398	3.000 / 2.417	2.000 / 1.611

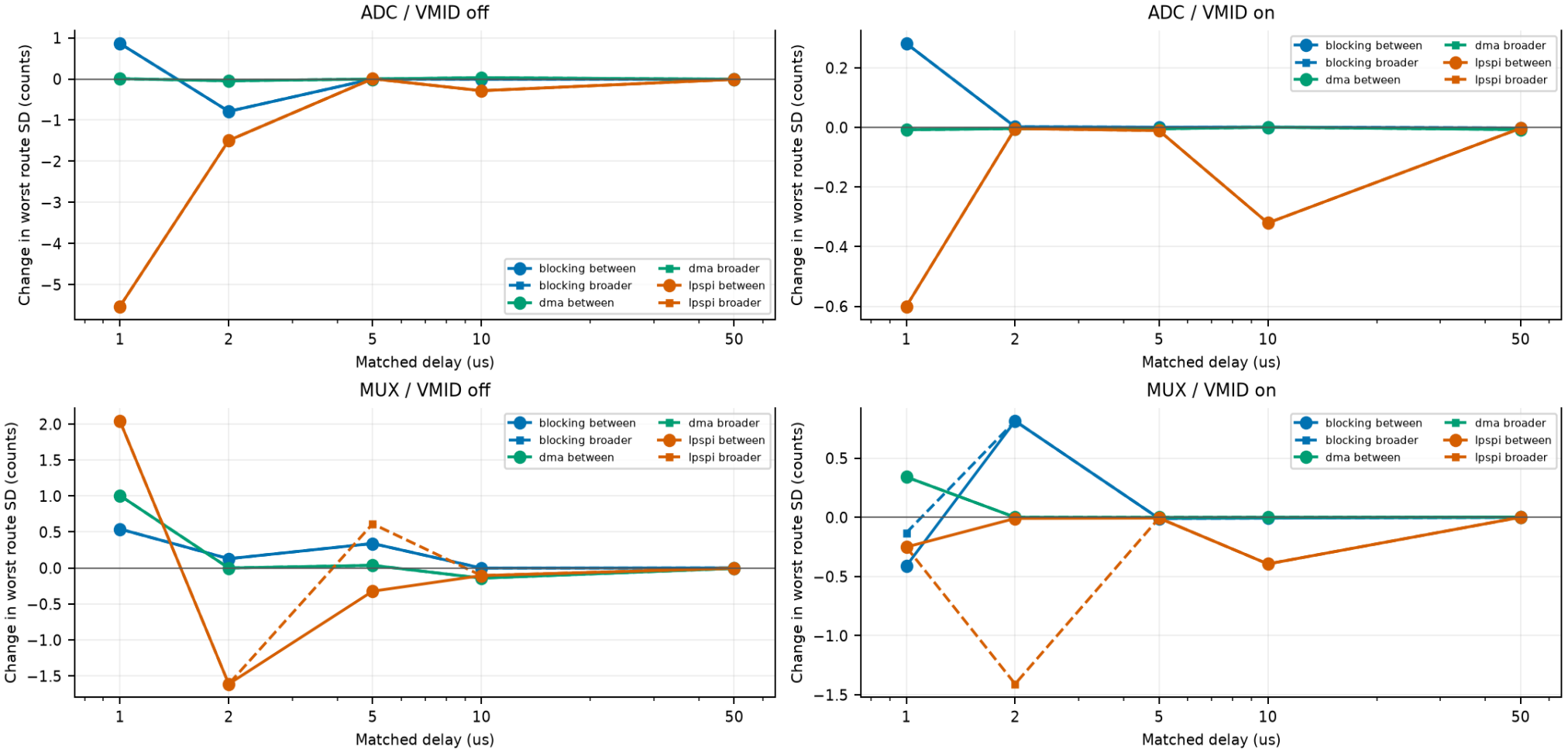
Nominal SPI clocking doubles from 64 to 128 us/sweep with VMID off, and from 134 to 268 us/sweep with VMID on. Total period includes settling and software overhead, so rate does not necessarily double at 16 MHz.

Speed change versus Yesterday's shared-MUX/burst firmware

Only 16 MHz settings with sensor = VMID settling are compared. Same scan-mode semantics are matched.



Quiet noise change versus Yesterday's shared-MUX/burst firmware



Positive change means higher noise. Solid lines: common between-burst windows; dashed lines: broader 50 ms windows. Each worst value is selected independently over routes. Both methods reanalyze the historical raw data. Yesterday's measurement windows were 2 s versus 15 s today; peak-to-peak extremes are not equal-length comparisons.

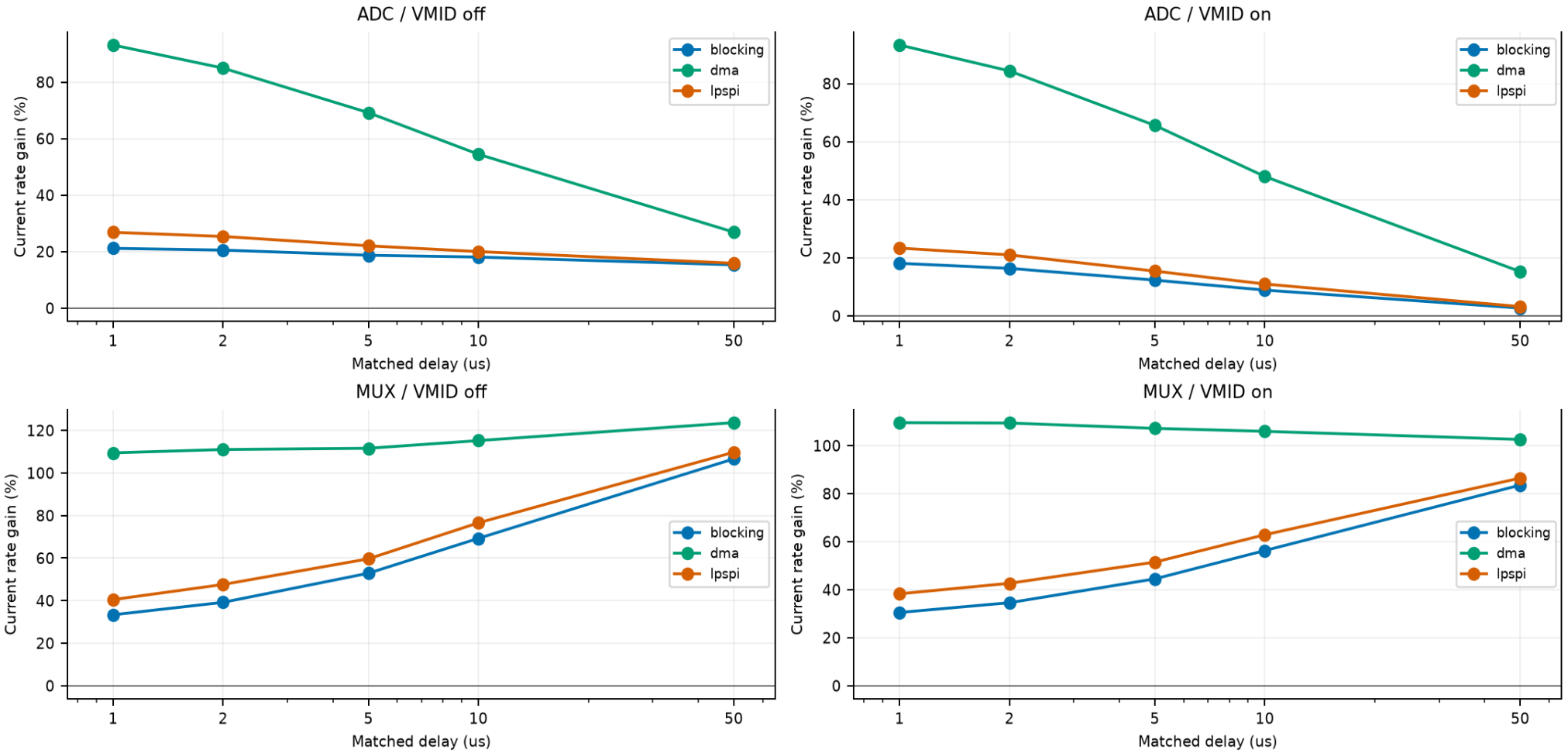
LPSPI comparison table - Yesterday's shared-MUX/burst firmware

Mode	VMID	us	Old rate	Current rate	Gain %	Old SD c/mV	Current SD c/mV	Old p-p c/mV	Current p-p c/mV	Old quiet reps
ADC	off	1	8,333.3	9,534.6	14.41	6.019 / 4.850	0.489 / 0.394	15.000 / 12.085	3.000 / 2.417	3/3
ADC	off	2	7,462.7	8,412.4	12.73	1.988 / 1.602	0.494 / 0.398	6.000 / 4.834	2.000 / 1.611	3/3
ADC	off	5	5,681.8	6,217.0	9.42	0.493 / 0.398	0.498 / 0.402	1.000 / 0.806	2.000 / 1.611	3/3
ADC	off	10	4,062.5	4,332.0	6.63	0.782 / 0.630	0.497 / 0.400	2.000 / 1.611	2.000 / 1.611	3/3
ADC	off	50	1,241.5	1,264.5	1.85	0.500 / 0.403	0.490 / 0.395	1.000 / 0.806	1.000 / 0.806	3/3
ADC	on	1	4,048.7	4,636.4	14.51	1.092 / 0.880	0.492 / 0.396	3.000 / 2.417	2.000 / 1.611	3/3
ADC	on	2	3,636.4	4,115.2	13.17	0.500 / 0.403	0.496 / 0.399	1.000 / 0.806	2.000 / 1.611	3/3
ADC	on	5	2,786.4	3,059.4	9.80	0.500 / 0.403	0.489 / 0.394	1.000 / 0.806	2.000 / 1.611	3/3
ADC	on	10	2,005.2	2,142.0	6.82	0.820 / 0.660	0.499 / 0.402	3.000 / 2.417	1.000 / 0.806	3/3
ADC	on	50	618.1	630.2	1.96	0.499 / 0.402	0.496 / 0.400	1.000 / 0.806	1.000 / 0.806	3/3
MUX	off	1	8,928.4	10,561.5	18.29	1.740 / 1.402	3.782 / 3.047	7.000 / 5.640	17.000 / 13.696	3/3
MUX	off	2	8,386.8	9,904.7	18.10	2.109 / 1.699	0.498 / 0.401	7.000 / 5.640	2.000 / 1.611	3/3
MUX	off	5	7,142.9	8,132.6	13.86	0.820 / 0.661	0.497 / 0.401	3.000 / 2.417	2.000 / 1.611	3/3
MUX	off	10	5,714.3	6,374.0	11.55	0.602 / 0.485	0.497 / 0.400	2.000 / 1.611	2.000 / 1.611	3/3
MUX	off	50	2,200.0	2,288.9	4.04	0.500 / 0.403	0.497 / 0.400	1.000 / 0.806	2.000 / 1.611	3/3
MUX	on	1	4,329.0	5,198.4	20.08	0.751 / 0.605	0.500 / 0.403	3.000 / 2.417	2.000 / 1.611	3/3
MUX	on	2	4,094.2	4,852.4	18.52	0.508 / 0.409	0.498 / 0.401	2.000 / 1.611	2.000 / 1.611	3/3
MUX	on	5	3,484.5	4,016.1	15.26	0.500 / 0.403	0.493 / 0.398	1.000 / 0.806	2.000 / 1.611	3/3
MUX	on	10	2,803.5	3,144.3	12.15	0.891 / 0.718	0.497 / 0.400	3.000 / 2.417	2.000 / 1.611	3/3
MUX	on	50	1,091.8	1,138.9	4.32	0.500 / 0.403	0.500 / 0.403	1.000 / 0.806	1.000 / 0.806	3/3

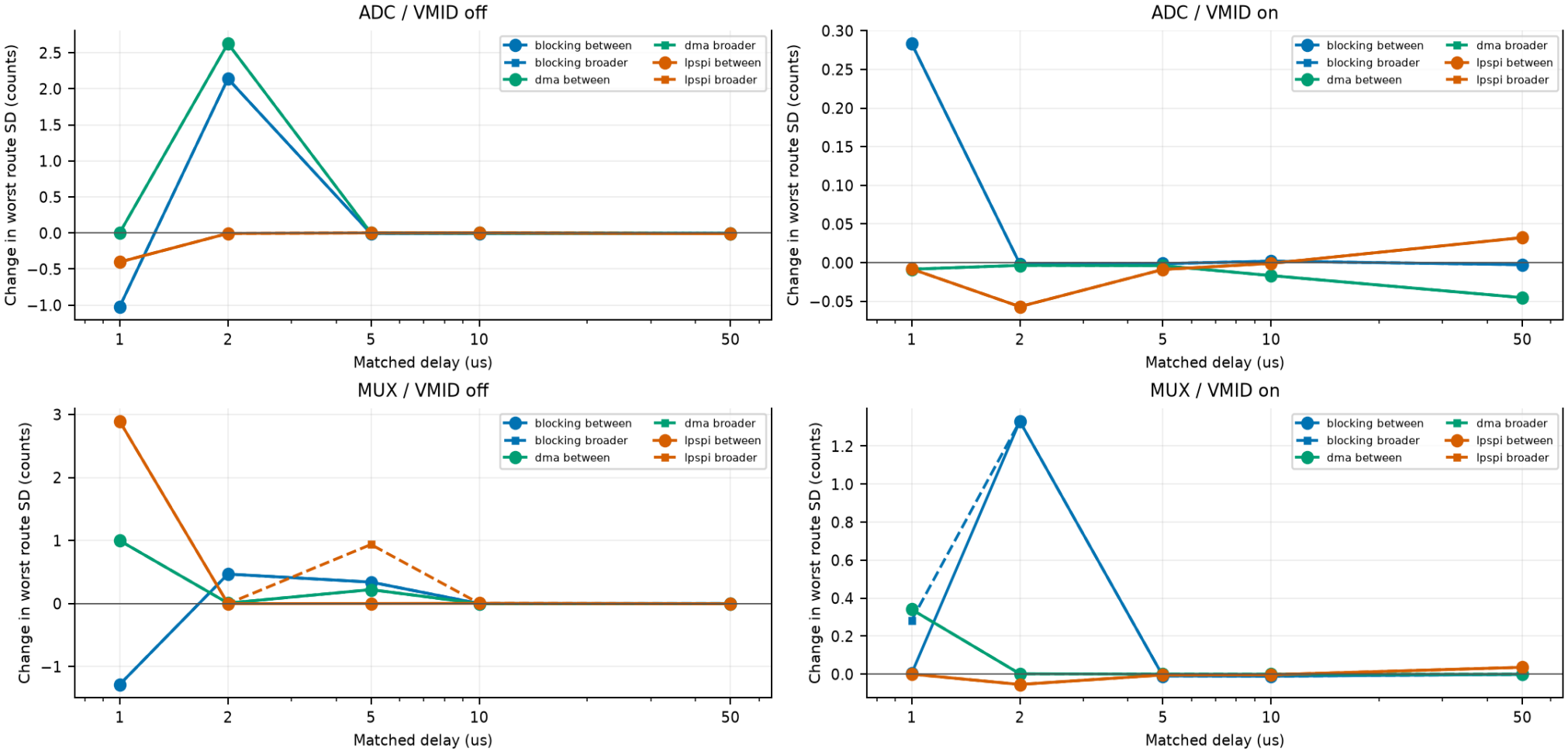
All three engines and every matched physical route are included in firmware_comparisons.csv and channel_comparisons.csv. Unqualified old noise records are excluded rather than replaced with zeros. Sequential-session environmental drift remains part of this comparison.

Speed change versus Yesterday's original sequential firmware

Only 16 MHz settings with sensor = VMID settling are compared. The old scanorder mux means sequential arrays, equivalent to today's ADC mode. Today's MUX results compare against that sequential reference and include the change in scan behavior.



Quiet noise change versus Yesterday's original sequential firmware



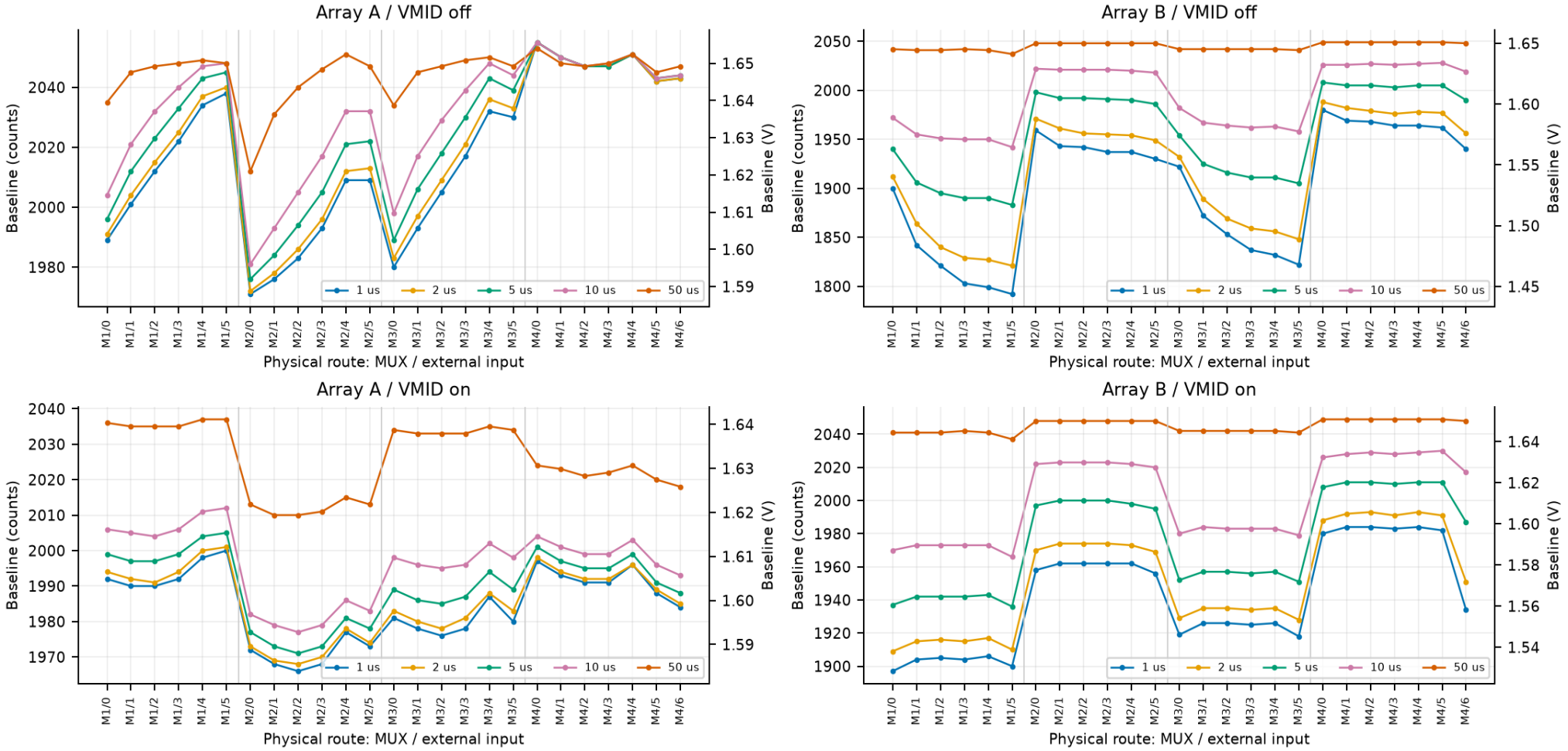
Positive change means higher noise. Solid lines: common between-burst windows; dashed lines: broader 50 ms windows. Each worst value is selected independently over routes. Both methods reanalyze the historical raw data. Both sessions have 15 s measurement windows, but retained quiet durations and sample counts differ.

LPSPI comparison table - Yesterday's original sequential firmware

Mode	VMID	us	Old rate	Current rate	Gain %	Old SD c/mV	Current SD c/mV	Old p-p c/mV	Current p-p c/mV	Old quiet reps
ADC	off	1	7,518.1	9,534.6	26.82	0.891 / 0.718	0.489 / 0.394	8.000 / 6.445	3.000 / 2.417	3/3
ADC	off	2	6,711.4	8,412.4	25.34	0.500 / 0.403	0.494 / 0.398	2.000 / 1.611	2.000 / 1.611	3/3
ADC	off	5	5,094.1	6,217.0	22.04	0.497 / 0.401	0.498 / 0.402	2.000 / 1.611	2.000 / 1.611	3/3
ADC	off	10	3,610.1	4,332.0	20.00	0.494 / 0.398	0.497 / 0.400	2.000 / 1.611	2.000 / 1.611	3/3
ADC	off	50	1,091.4	1,264.5	15.86	0.500 / 0.403	0.490 / 0.395	1.000 / 0.806	1.000 / 0.806	3/3
ADC	on	1	3,759.4	4,636.4	23.33	0.500 / 0.403	0.492 / 0.396	2.000 / 1.611	2.000 / 1.611	3/3
ADC	on	2	3,401.4	4,115.2	20.99	0.553 / 0.445	0.496 / 0.399	3.000 / 2.417	2.000 / 1.611	3/3
ADC	on	5	2,651.3	3,059.4	15.39	0.498 / 0.401	0.489 / 0.394	1.000 / 0.806	2.000 / 1.611	3/3
ADC	on	10	1,930.5	2,142.0	10.95	0.500 / 0.403	0.499 / 0.402	1.000 / 0.806	1.000 / 0.806	3/3
ADC	on	50	610.8	630.2	3.17	0.464 / 0.374	0.496 / 0.400	1.000 / 0.806	1.000 / 0.806	3/3
MUX	off	1	7,518.1	10,561.5	40.48	0.891 / 0.718	3.782 / 3.047	8.000 / 6.445	17.000 / 13.696	3/3
MUX	off	2	6,711.4	9,904.7	47.58	0.500 / 0.403	0.498 / 0.401	2.000 / 1.611	2.000 / 1.611	3/3
MUX	off	5	5,094.1	8,132.6	59.65	0.497 / 0.401	0.497 / 0.401	2.000 / 1.611	2.000 / 1.611	3/3
MUX	off	10	3,610.1	6,374.0	76.56	0.494 / 0.398	0.497 / 0.400	2.000 / 1.611	2.000 / 1.611	3/3
MUX	off	50	1,091.4	2,288.9	109.73	0.500 / 0.403	0.497 / 0.400	1.000 / 0.806	2.000 / 1.611	3/3
MUX	on	1	3,759.4	5,198.4	38.28	0.500 / 0.403	0.500 / 0.403	2.000 / 1.611	2.000 / 1.611	3/3
MUX	on	2	3,401.4	4,852.4	42.66	0.553 / 0.445	0.498 / 0.401	3.000 / 2.417	2.000 / 1.611	3/3
MUX	on	5	2,651.3	4,016.1	51.48	0.498 / 0.401	0.493 / 0.398	1.000 / 0.806	2.000 / 1.611	3/3
MUX	on	10	1,930.5	3,144.3	62.87	0.500 / 0.403	0.497 / 0.400	1.000 / 0.806	2.000 / 1.611	3/3
MUX	on	50	610.8	1,138.9	86.46	0.464 / 0.374	0.500 / 0.403	1.000 / 0.806	1.000 / 0.806	3/3

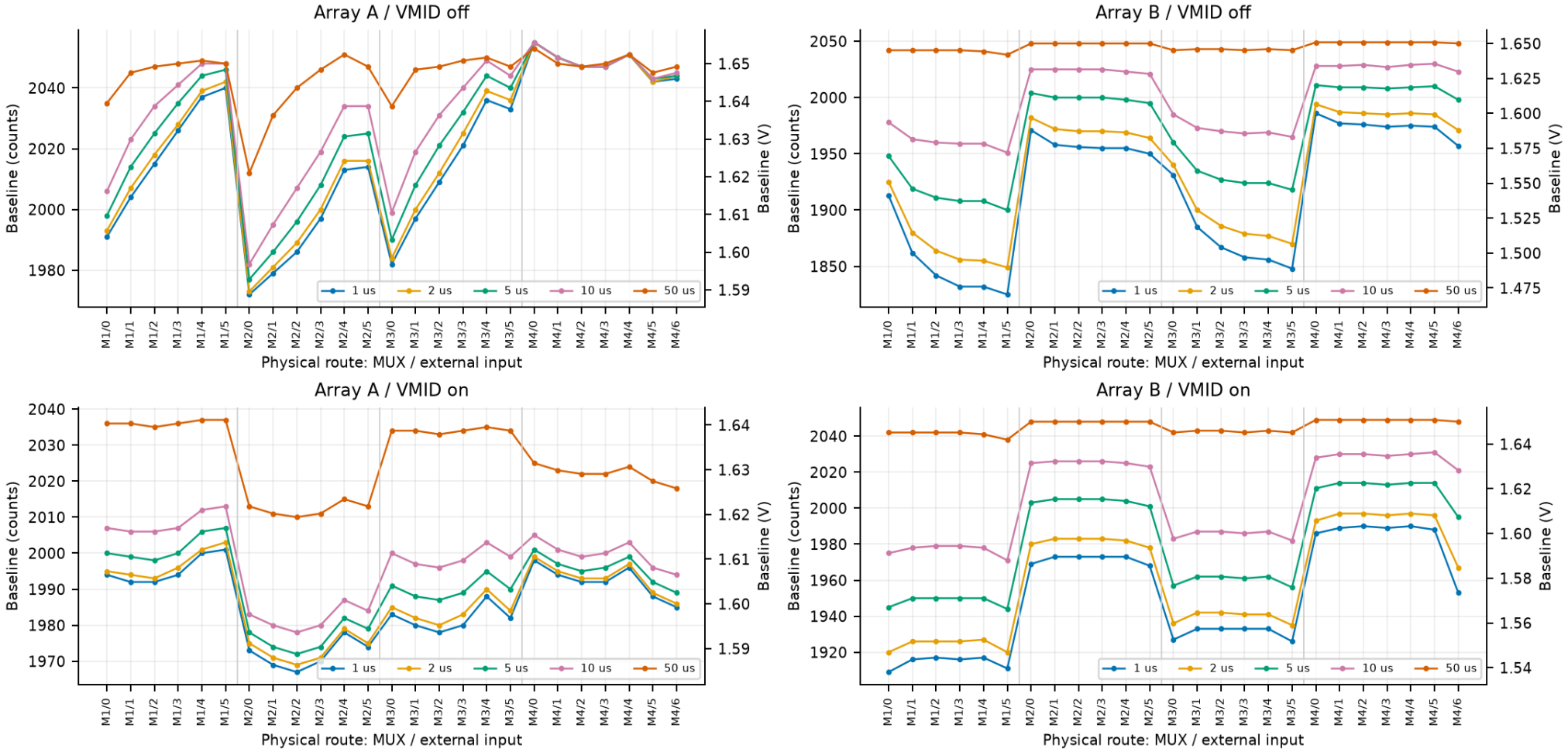
All three engines and every matched physical route are included in firmware_comparisons.csv and channel_comparisons.csv. Unqualified old noise records are excluded rather than replaced with zeros. Sequential-session environmental drift remains part of this comparison.

Every channel baseline - ADC / blocking



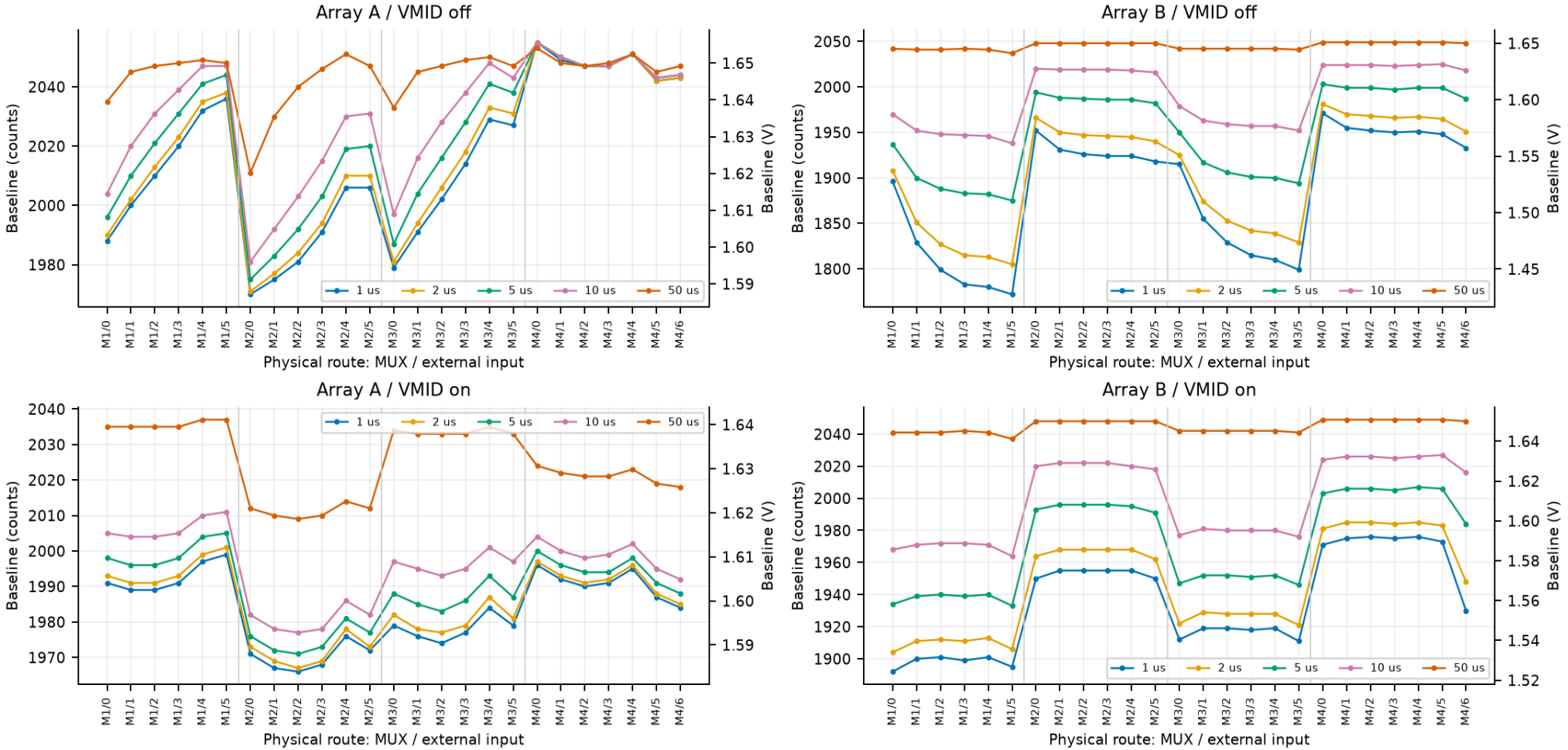
16 MHz; sensor = VMID settling. Each point is the median quiet baseline for that physical route, aggregated by median over qualified repetitions. Lines connect discrete channels for comparison and do not imply spatial interpolation. Left axis counts; right axis volts.

Every channel baseline - ADC / dma



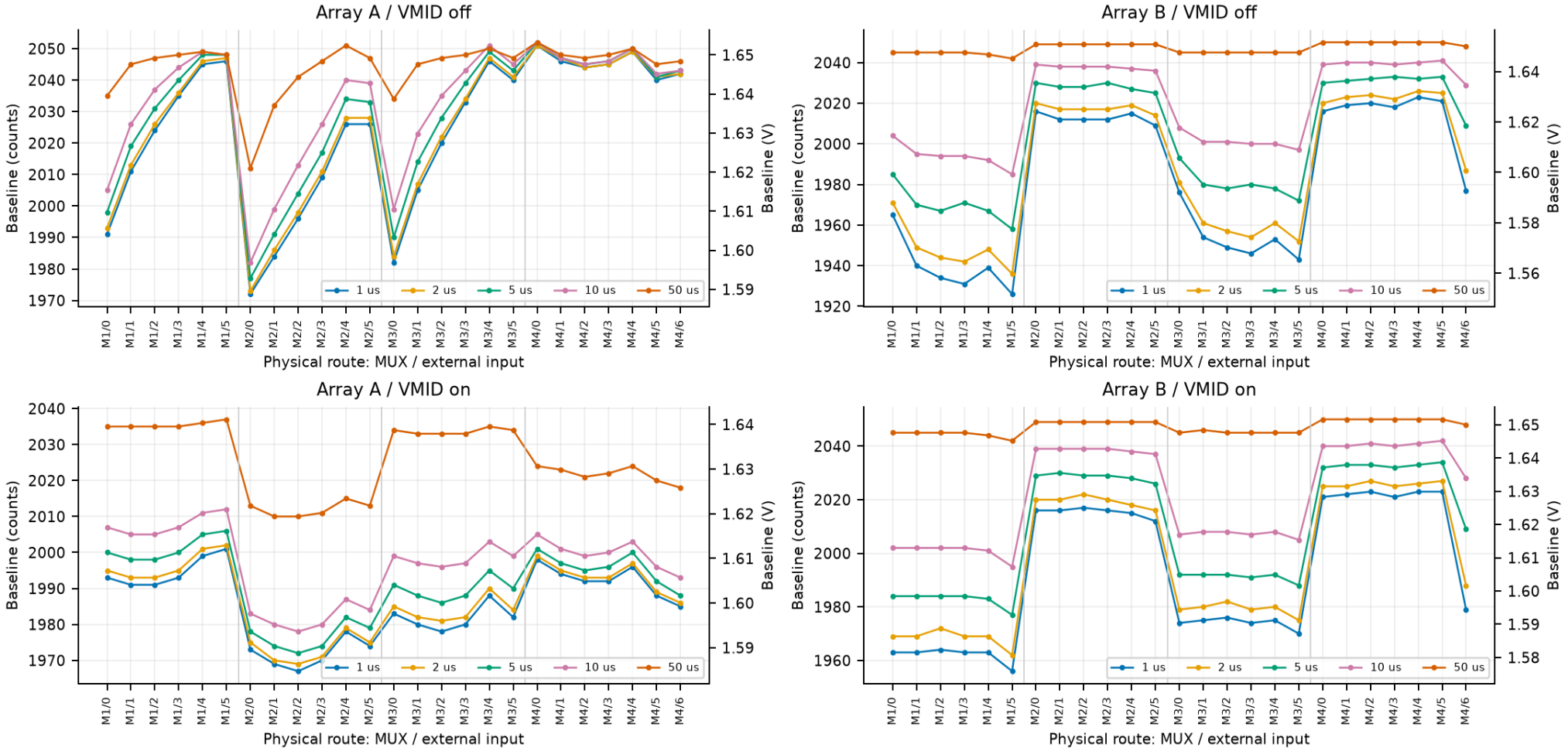
16 MHz; sensor = VMID settling. Each point is the median quiet baseline for that physical route, aggregated by median over qualified repetitions. Lines connect discrete channels for comparison and do not imply spatial interpolation. Left axis counts; right axis volts.

Every channel baseline - ADC / lpspi



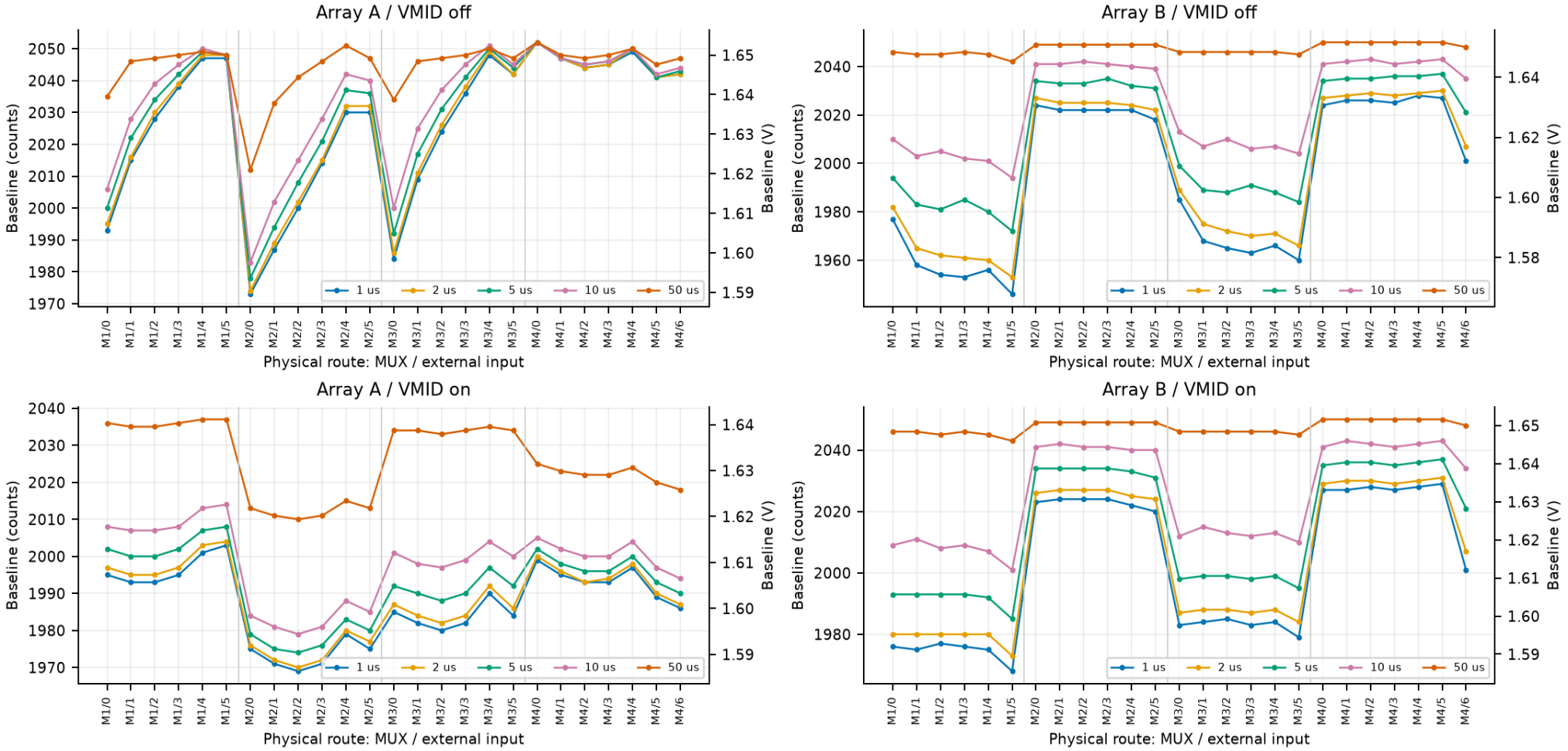
16 MHz; sensor = VMID settling. Each point is the median quiet baseline for that physical route, aggregated by median over qualified repetitions. Lines connect discrete channels for comparison and do not imply spatial interpolation. Left axis counts; right axis volts.

Every channel baseline - MUX / blocking



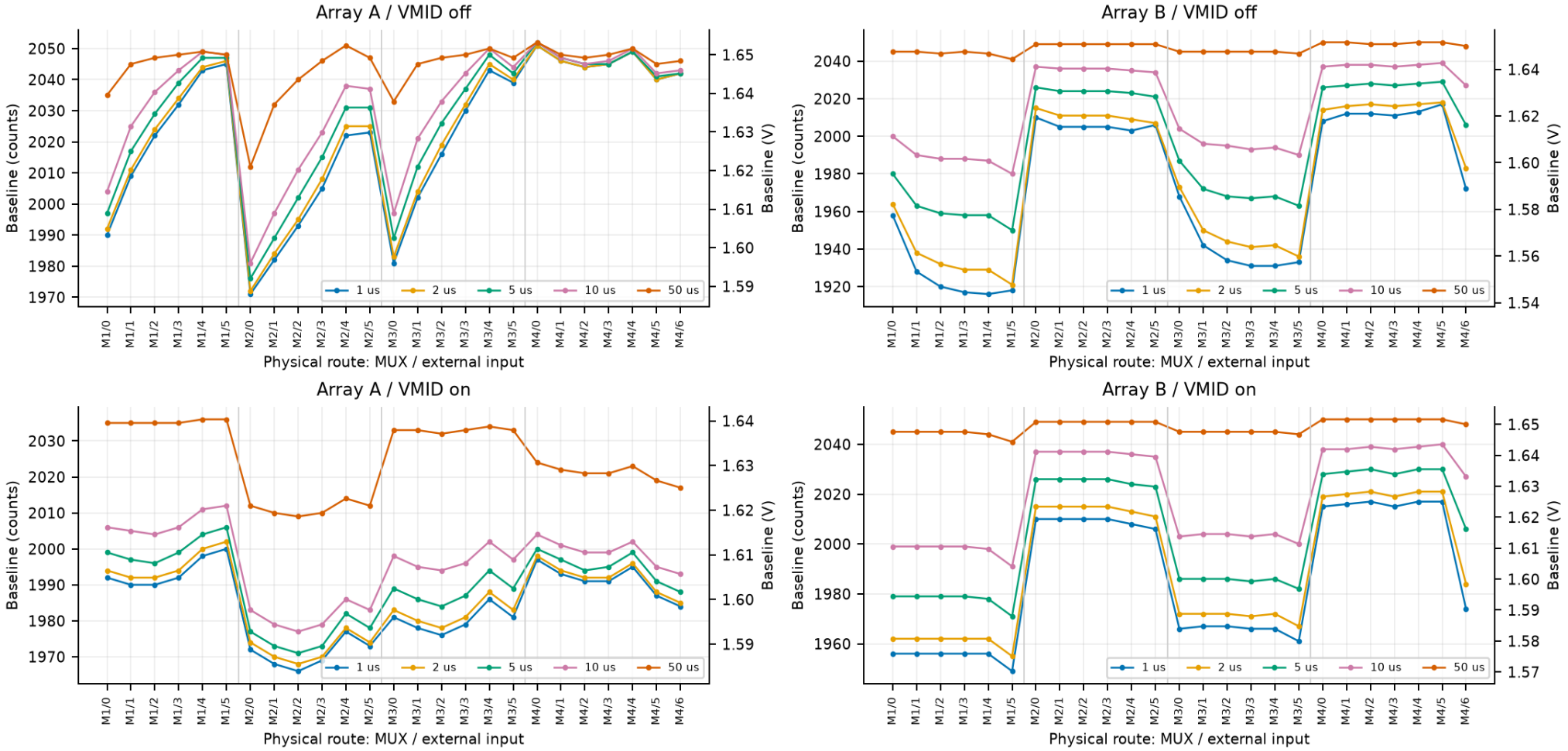
16 MHz; sensor = VMID settling. Each point is the median quiet baseline for that physical route, aggregated by median over qualified repetitions. Lines connect discrete channels for comparison and do not imply spatial interpolation. Left axis counts; right axis volts.

Every channel baseline - MUX / dma



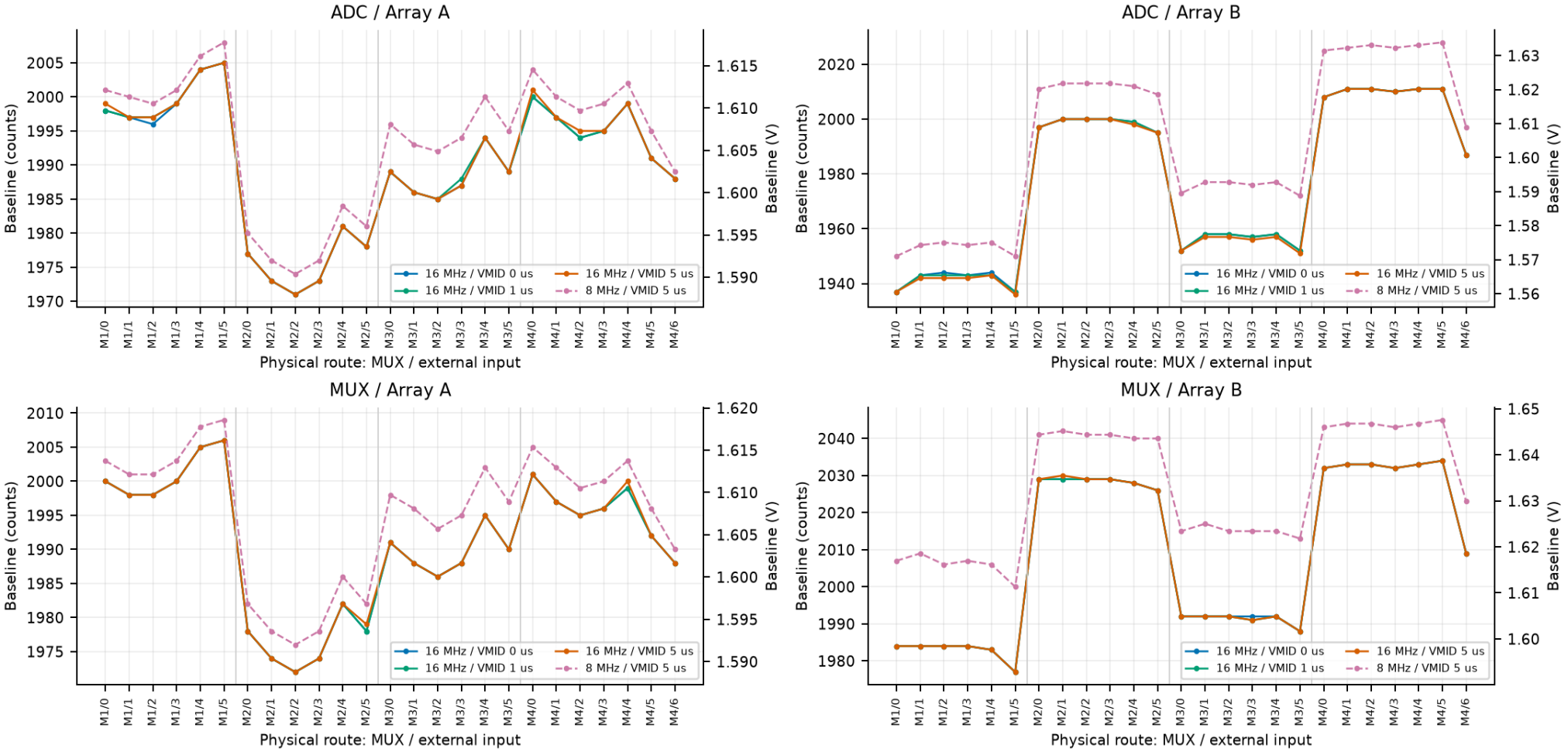
16 MHz; sensor = VMID settling. Each point is the median quiet baseline for that physical route, aggregated by median over qualified repetitions. Lines connect discrete channels for comparison and do not imply spatial interpolation. Left axis counts; right axis volts.

Every channel baseline - MUX / lpspi



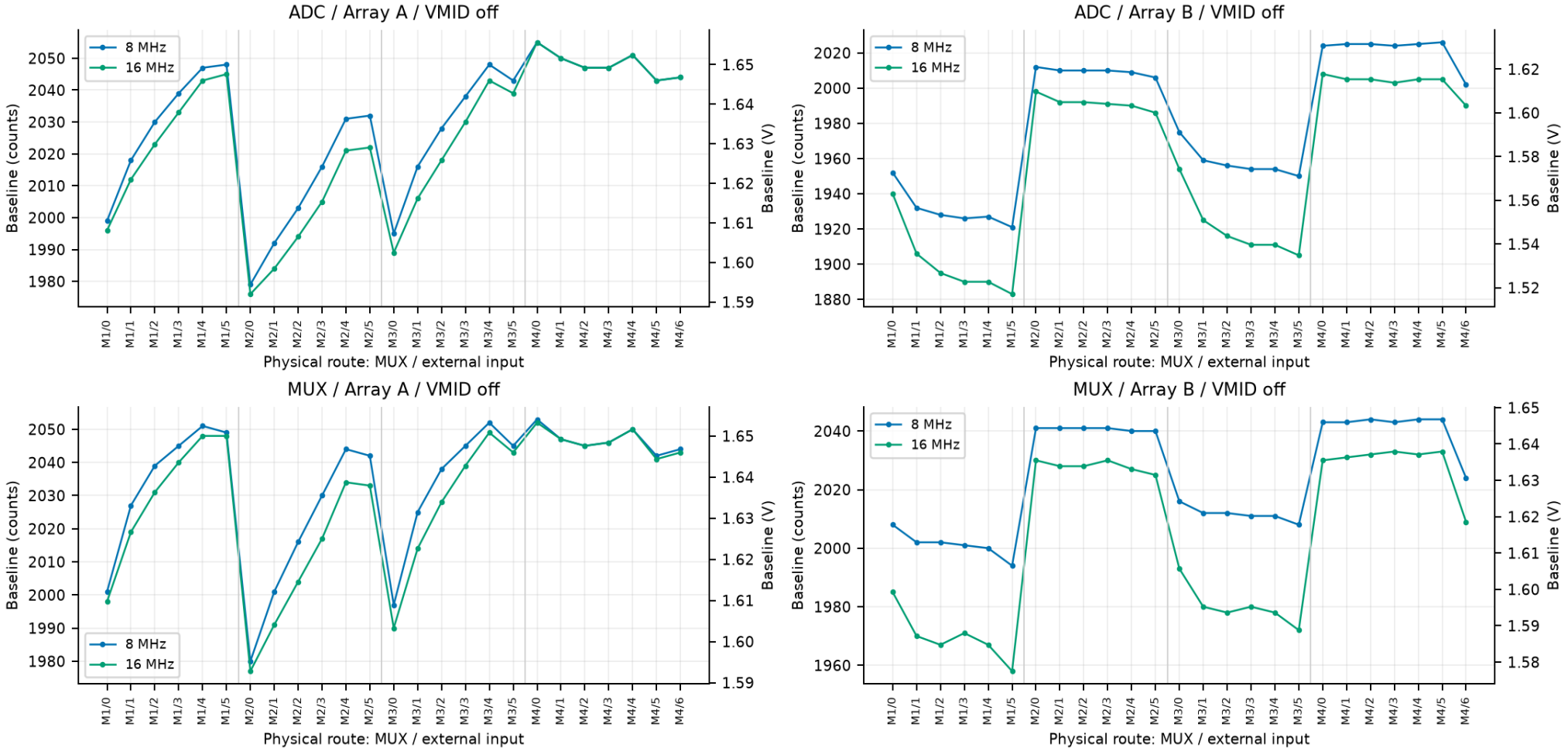
16 MHz; sensor = VMID settling. Each point is the median quiet baseline for that physical route, aggregated by median over qualified repetitions. Lines connect discrete channels for comparison and do not imply spatial interpolation. Left axis counts; right axis volts.

Every channel baseline - VMID and clock experiments / blocking



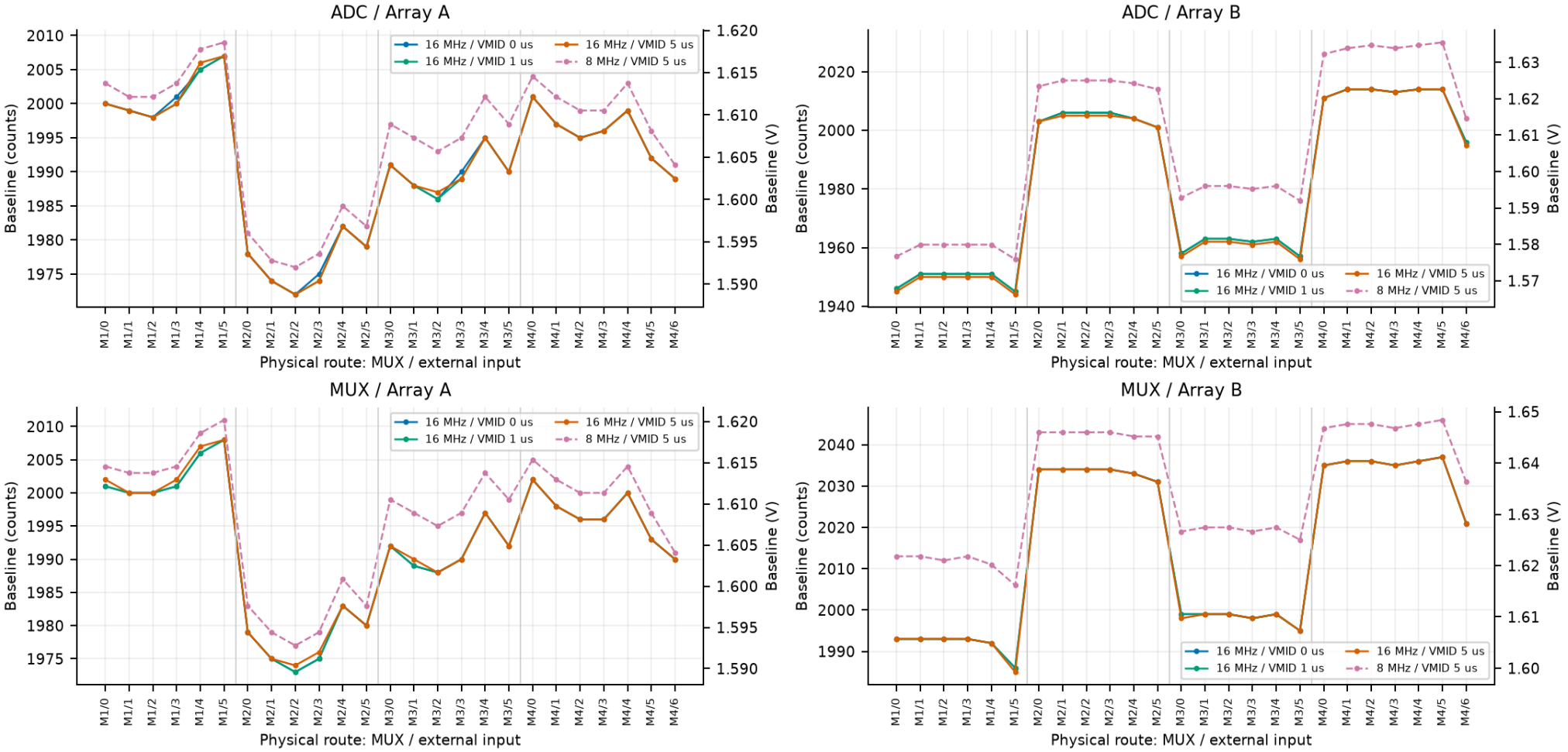
Sensor settling fixed at 5 us; VMID on. All four experimental settings are plotted in physical channel order.

Every channel baseline - 8/16 MHz with VMID off / blocking



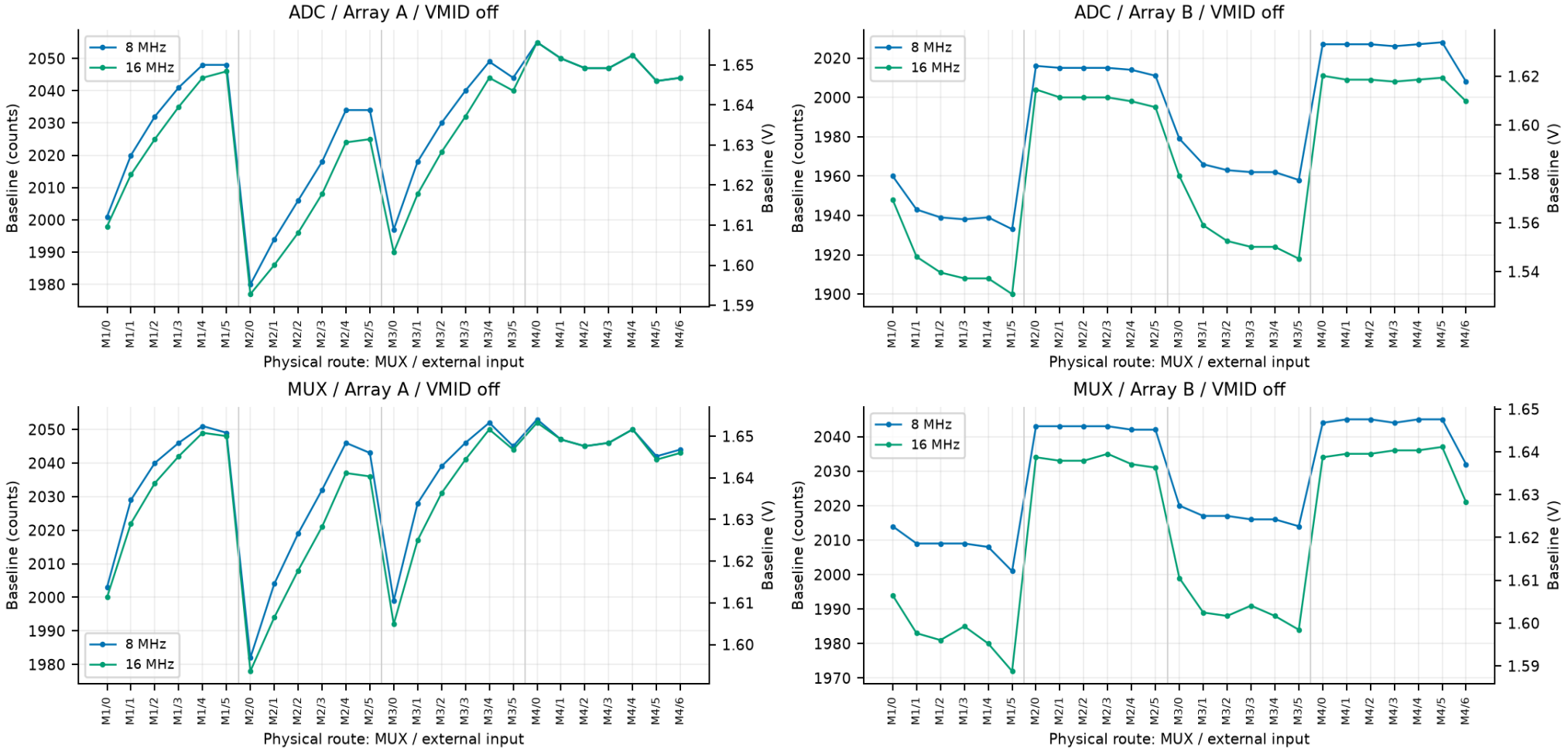
Sensor settling = VMID settling = 5 us; VMID off. Both requested SPI clocks are shown for every physical channel in both scan modes.

Every channel baseline - VMID and clock experiments / dma



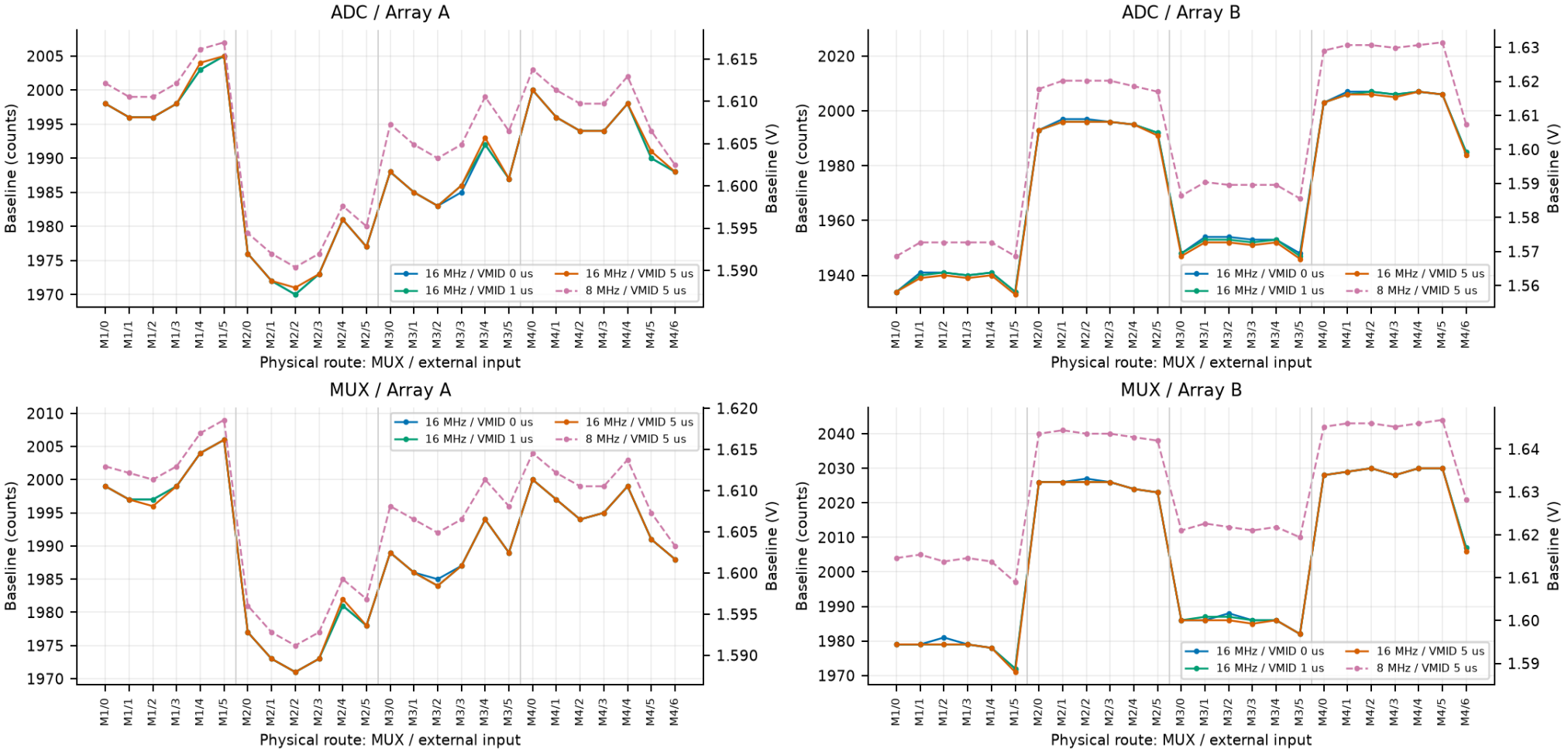
Sensor settling fixed at 5 us; VMID on. All four experimental settings are plotted in physical channel order.

Every channel baseline - 8/16 MHz with VMID off / dma



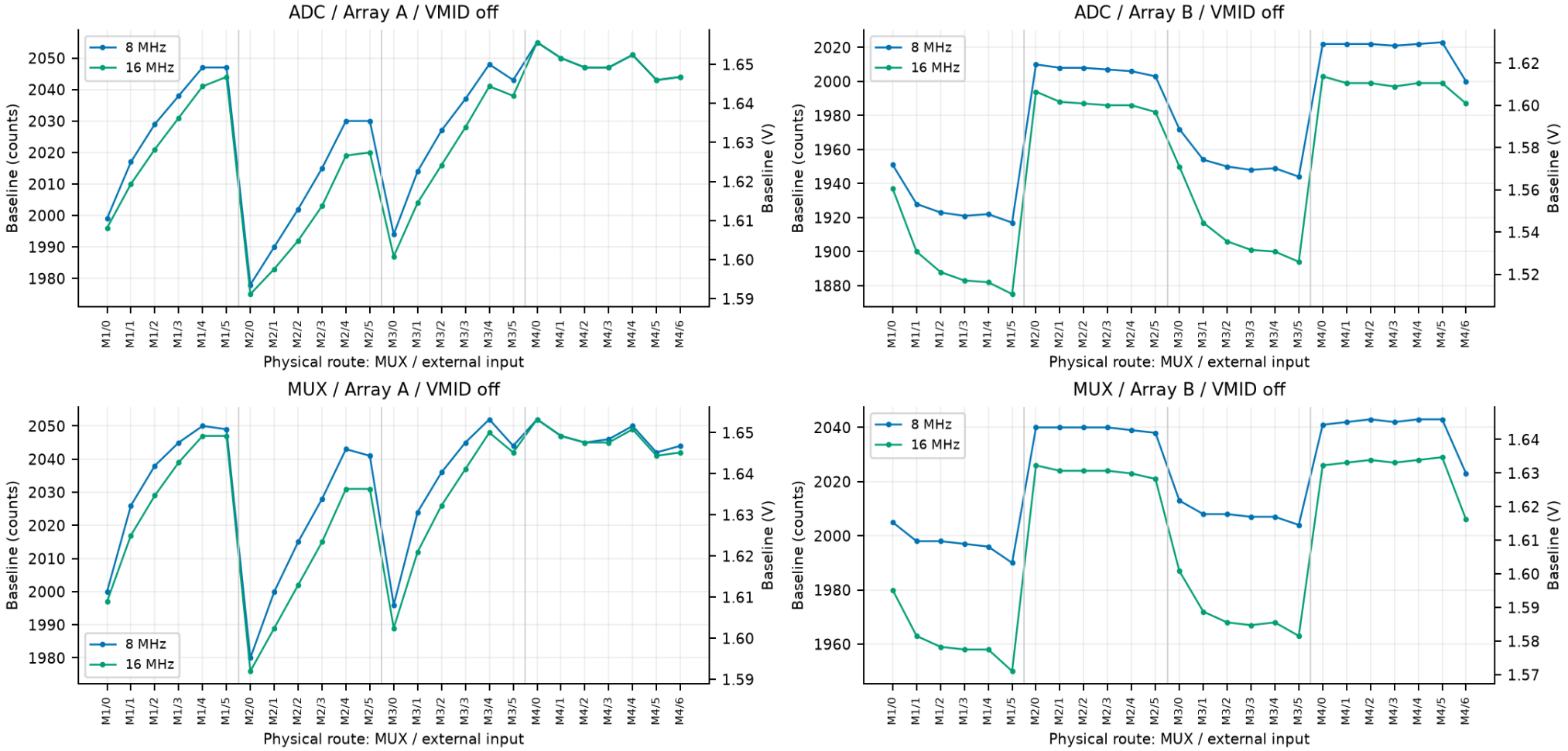
Sensor settling = VMID settling = 5 us; VMID off. Both requested SPI clocks are shown for every physical channel in both scan modes.

Every channel baseline - VMID and clock experiments / lpspi



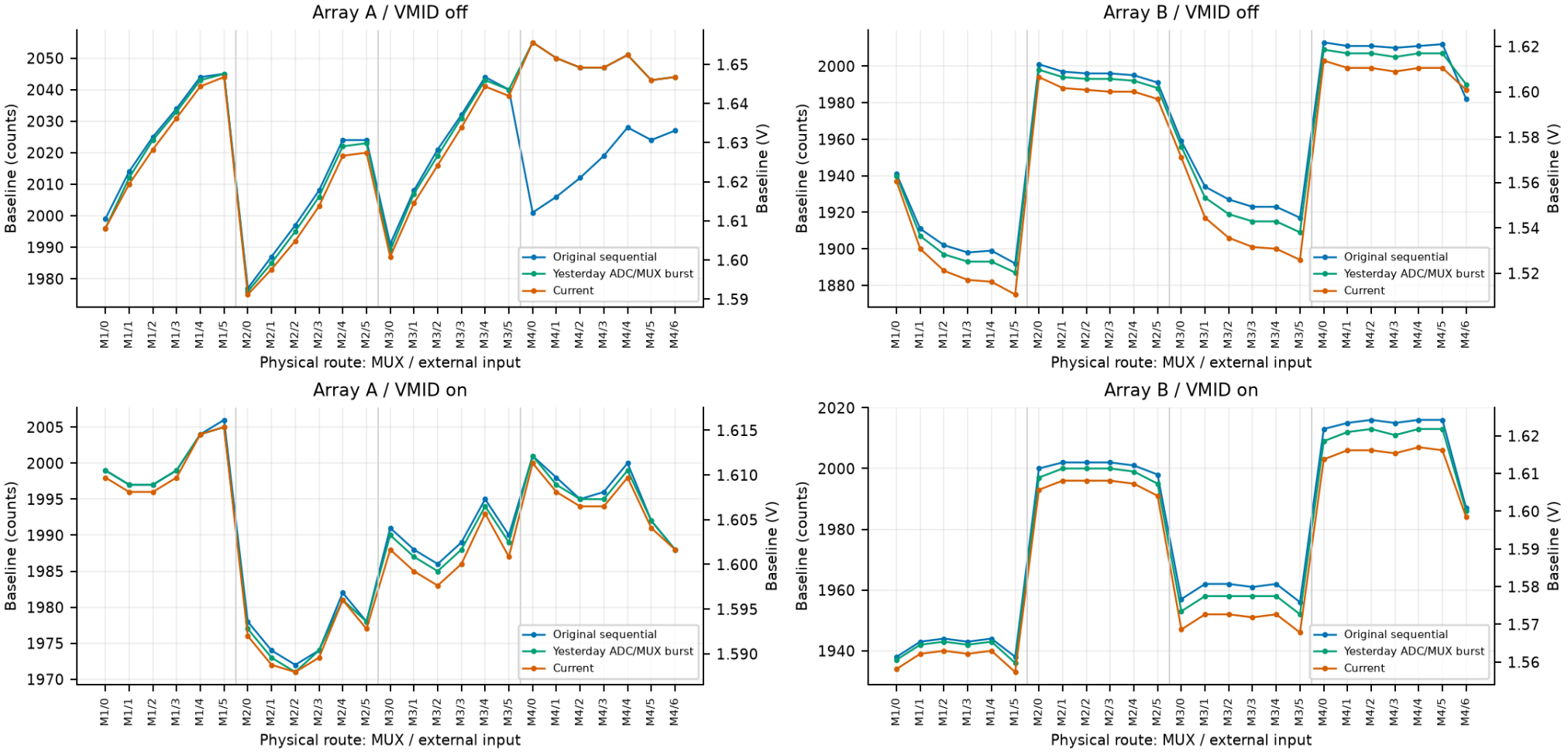
Sensor settling fixed at 5 us; VMID on. All four experimental settings are plotted in physical channel order.

Every channel baseline - 8/16 MHz with VMID off / lpspi



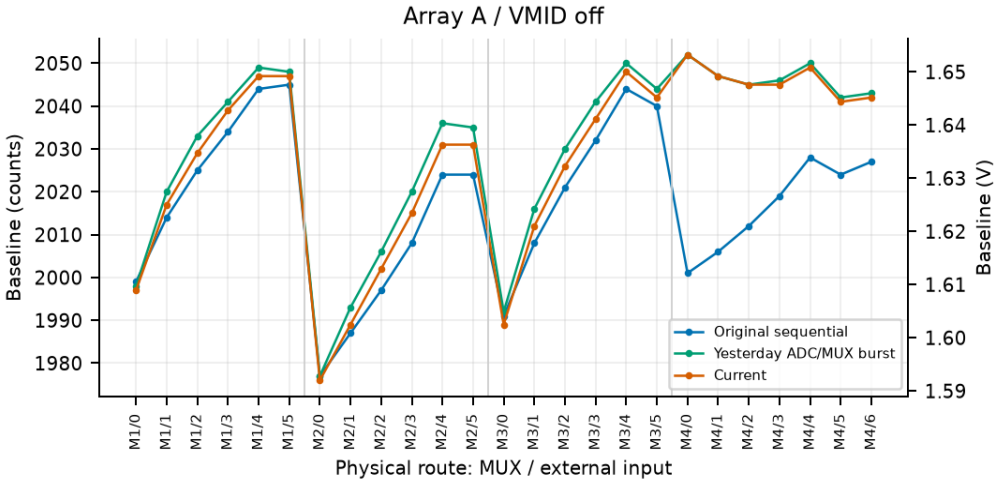
Sensor settling = VMID settling = 5 us; VMID off. Both requested SPI clocks are shown for every physical channel in both scan modes.

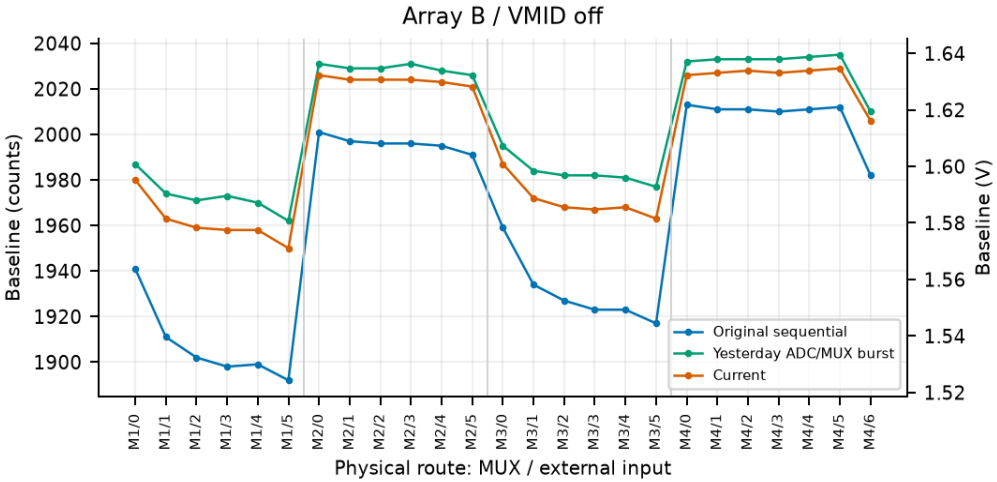
Every channel baseline - firmware history / LPSPI / ADC

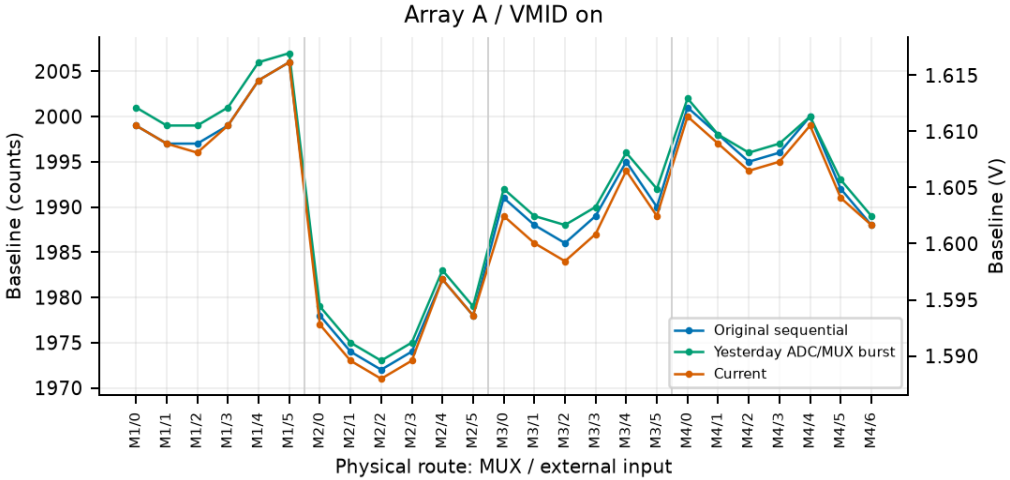


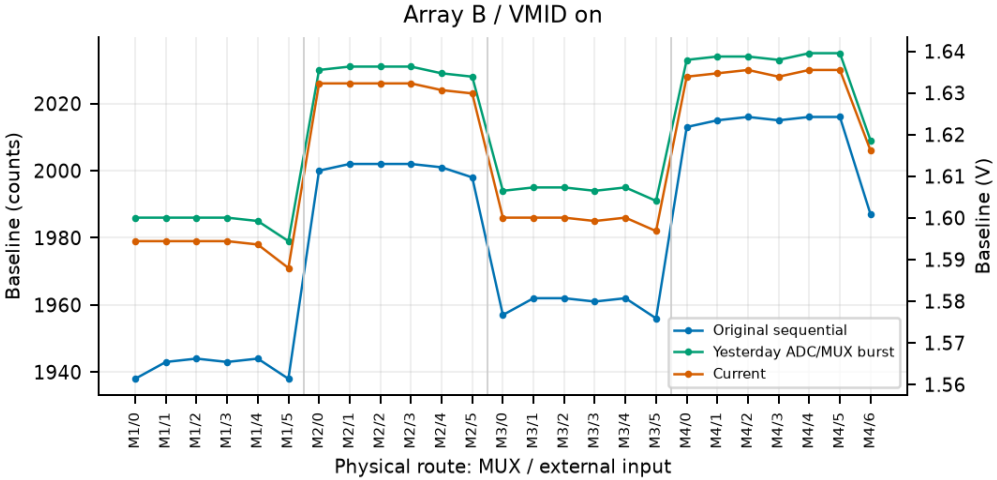
Sensor = VMID settling = 5 us; 16 MHz. Original firmware is sequential for both comparisons. Yesterday's burst firmware matches the current scan mode. Every delay/engine route comparison is saved in CSV.

Every channel baseline - firmware history / LPSPI / MUX



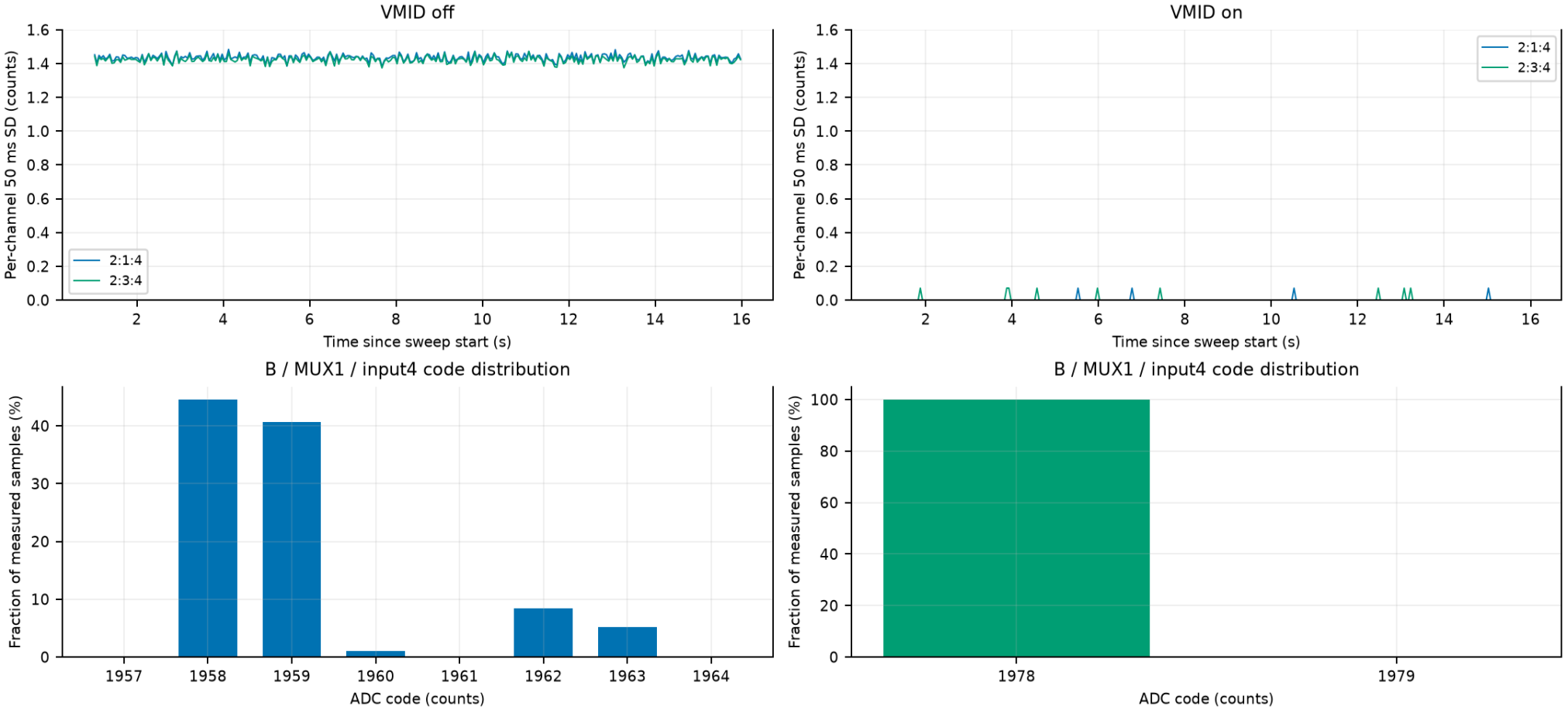






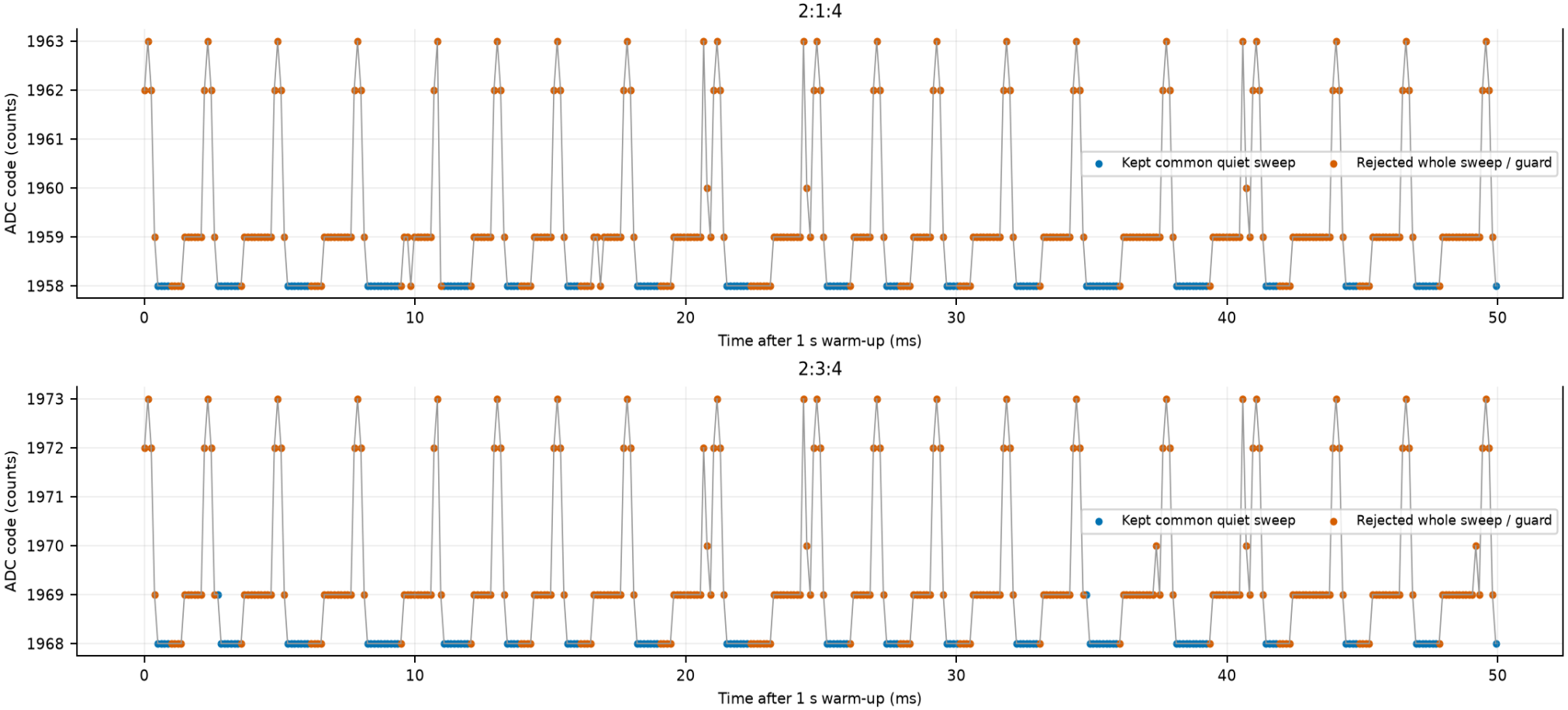
Sensor = VMID settling = 5 us; 16 MHz. Original firmware is sequential for both comparisons. Yesterday's burst firmware matches the current scan mode. Every delay/engine route comparison is saved in CSV.

Fast coherent excursions hidden inside the 50 ms envelope



Illustrative first repetition, current MUX/LPSPI, 16 MHz, sensor=VMID settling=5 us. Routes 2:1:4 and 2:3:4 are B/MUX1/input4 and B/MUX3/input4 (both 1 MOhm bias). The earlier 50 ms mask retains these windows, but a finer trace reveals separated short excursions. Histograms include every measured sample. The broader-window metric includes these events; primary noise uses between-burst windows. This plot does not identify their electrical cause.

Common between-burst time windows - short trace



Same first VMID-off capture, first 50 ms after warm-up. Excursion intervals are selected from separated secondary states moving coherently on at least two routes. A whole sweep and one neighboring sweep on each side are rejected for every channel together. A point can be rejected because other channels trigger the common window; there is no independent clipping of this route. Full packed masks and detector states accompany every production capture.

Quiet-period selection and statistical definitions

Exclude the first 1 s of each run. In complete 50 ms bins, calculate each channel's sample SD (ddof=1), then take its median within each of the eight array/MUX groups. For each group, use threshold max(0.6 counts, P20+0.15, lower-40% median + 4 x 1.4826 x MAD). Reject any bin exceeding any group threshold, plus its adjacent bins. Apply the same retained time mask to all 50 channels.

This first stage identifies longer envelope bursts. A second stage detects shorter coherent excursions. A one-code modal state containing >=60% of coarse-window samples, or an adjacent two-code state containing >=80%, anchors detection. Allow one count outside that state. An excursion direction must contain >=1% of samples beyond the allowed edge and a separated histogram peak >=0.5% of samples and >=3 times the edge density. At least two routes must depart in the same sweep; reject that entire sweep and one neighboring sweep each side.

The same final sweep mask applies to all 50 channels. There is no independent channel clipping, smoothing or detrending. Unimodal Gaussian tails and isolated channel variation remain. Charger switching was not recorded, so the detector classifies coherent excursions without proving their source. Both the broader first-stage metrics and final conditional metrics remain available. A record qualifies with >=1 s and >=100 retained frames. Fine quiet seconds sum retained sweep intervals capped at the median sweep period, excluding credit for missing frames.

Baseline: median of retained samples per physical route. Temporal noise SD: sample SD about that route's retained mean. Peak-to-peak: retained max-min. Per-setting metrics: median of the three per-route repetition metrics. Baseline spread: max-min across those channel baselines; baseline population SD across channels is saved separately. Neither baseline spread nor its population SD is temporal noise.

Counts to volts uses the nominal reference, without calibration or reference-rail measurement. A reported zero SD means every retained digital code was identical; it does not establish zero analog noise. Peak-to-peak depends on observation duration and the number of sampled extremes. Repetition min/max fields show variation between captures.

In MUX mode, both banks share the configured settling wait, but ADC B is read after ADC A. Bank B therefore continues settling during the A batch. Equal delay settings across modes do not guarantee equal analog dwell for every route; baseline and noise comparisons retain those scheduling effects.

Dataset	Production records	Qualified quiet records	Measurement s
current	252	252	15
yesterday_shared	180	180	2
yesterday_original	90	90	15

Timing limits and optimization evidence

Steady-state current ADC mode has 14 sensor settling windows/sweep; MUX mode has 7. VMID on adds 14/7 VMID waits. VMID off has 64 conversion words and 14 burst batches; VMID on has 134 words and 28 batches. The preparation/stop parking waits are outside the steady-state model. Guard arithmetic preserves EN polarity and configured 200 ns turn-off/turn-on allowances.

Current firmware uses bounded LPSPI FIFO bursts and grouped GPIO writes. Measured residuals include GPIO/register work, batch setup, loop costs, interrupts and timestamp quantization; they are not exact per-channel aperture times. Independently aggregated component medians may not sum exactly to the median total. USB enqueue timing is not end-to-end host latency. Hardware edge timing, crosstalk and connected-sensor analog behavior require separate measurements.

Evidence, reproducibility and final board state

Current raw root: docs\testing\datasets\testboard-adc124\upgrade_disconnected_2026-10-14

Verified reference raw root: docs\testing\datasets\testboard-adc124\scan_modes_20261013T214607Z\production

Verified reference raw root: docs\testing\datasets\testboard-adc124\mux_production_2026-10-13

Current sources, production and diagnostic HEX images and SHA-256 identities are frozen under current_firmware/. Command logs, status before/after every capture, raw binary streams, independent parser checks, profiling summaries and lifecycle records accompany the data. Hashes identify the local builds; initial production identity was checked through its runtime revision and reported policies, without MCU image readback. The frozen diagnostic and restored production images are uploaded by this session.

Final lifecycle: complete_production_restored_verified. Production revision: adc124-vmid-settle-v1. Acquisition running: false. Original settings restored: array=both, scanorder=adc, spiengine=blocking, spi_clock_hz=16000000, mux_settle_us=5, vmid_settle_us=5, vmid_settle_mode=follow_sensor, vmid_between_groups=false.

File	Contents
capture_metrics.csv	Every raw capture: timing, transport counters, quiet qualification, estimates and diagnostic phases
summary.csv	Every setting: rates, timing budgets, quiet noise, baseline spread and repetition ranges
channel_repetitions.csv / channel_summary.csv	All 50 physical routes, in counts and volts; baseline, SD and peak-to-peak
firmware_comparisons.csv / channel_comparisons.csv	Matched-delay firmware speed/noise comparisons and route baseline shifts
mux_group_summary.csv / address_baseline_spread.csv	Within-MUX, across-lane and same-address baseline spreads
delay_fits.csv	Measured delay slopes/intercepts and expected steady-state wait counts
broader_window_summary.csv / broader_window_channel_summary.csv	The earlier 50 ms-window noise/baseline metrics, retained alongside between-burst results
firmware_broader_comparisons.csv	Matched-setting broader-window noise comparisons for both historical references
*quiet_mask.npz / *fine_quiet_states.json	Raw-frame-indexed common masks and reproducible separated-state detector parameters
figures/	All report figures and extra VMID-off 8/16 MHz baseline line plots
analysis.json / analysis_cache.json	Structured results and raw-hash-bound analysis cache

Reproduce with scripts/benchmark_adc124_upgrade.py, scripts/analyze_adc124_upgrade.py and scripts/report_adc124_upgrade.py. No firmware source changes, commit or push were made for this measurement session. The disconnected-array captures should be preserved as the reference for the later sensor-connected experiment.